## COMSATS University Islamabad, Pakistan

# EmoSense

**A project presented to**
**COMSATS Institute of Information Technology, Islamabad**

**In partial fulfillment**
**of the requirement for the degree of**

***Bachelor of Science in Computer Science (2021-2025)***

**By**


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# COMSATS University Islamabad, Sahiwal Campus

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**The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.**

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**Haseeb Shahid** **Ibad Elahi**

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# CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) "EmoSense" was developed by Haseeb Shahid (CIIT/FA21-BCS-054) and Ibad Elahi (CIIT/FA21-BCS-103) under the supervision of "Ms. Tahreem Saeed" and co supervisor "Ms. Hafsa Mehreen" and that in her opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in Computer Sciences.

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# Executive Summary

In public places, there is often a need for monitoring people and different activities going on, which can be referred later for many reasons including security. Appointing humans for this task involves many problems such as increased employee hiring, accuracy problems, trust, no proof for later use, and also the fact that a human can remember things till a certain time limit. Talking about the current security system, they use dumb still cameras with a continuous recording facility irrespective of the fact that any event may happen or not. Moreover, they are usually pointing at a specific user-defined locations so more than one camera are required to cover the entire region.

To prevent all these problems from prevailing, the CSCS is developed. It is a surveillance system, which provides solutions to many of these problems. It is a stand-alone application which doesn't require any computer to operate. It monitors different situations using a camera which can rotate intelligently based on sensor messages and captures the scene in the form of video or photos later reference as well.

Customizable Surveillance Control System (CSCS) is a surveillance system that can be assigned to a sensor type as in our case a heat sensor is used, it works accordingly, rotates the camera upon event detection and performs user defined actions like capturing video and stores them, for future use.

It is an embedded system consisting of Linux fox kit with embedded a running server application also a camera, USB storage device and a sensor node base station is attached with fox kit. LAN communication is used by users to download videos and to operate the system manually.

# Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor “Ms. Tahreem Saeed” and our co-Supervisor “Ms. Hafsa Mehreen”. Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

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# Abbreviations

| | |
|---|---|
| **API** | Application Programming Interface |
| **CS** | Computer Science |
| **SRS** | Software Require Specification |
| **SDD** | Software Design Description |
| **GUI** | Graphical User Interface |
| **AG** | Agile Model |
| **IDEs** | Integrated Development Environments |
| **NLP** | Natural language Processing |
| **PC** | Personal Computer |

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# Introduction

# 1. Introduction

In today’s digitally connected yet emotionally distant world, maintaining mental and emotional well-being has become more important—and more challenging—than ever before. Many individuals struggle to find the right support when dealing with stress, anxiety, or emotional turmoil. Traditional solutions, such as therapy or counseling, can be costly, intimidating, or inaccessible to those who need immediate help. As a result, people often face emotional challenges alone, without the tools or guidance to process their feelings effectively.

EmoSense is a platform designed to revolutionize emotional wellness through the power of artificial intelligence. It offers a supportive, user-friendly interface where individuals can engage in meaningful conversations with an intelligent chatbot trained to respond empathetically and help users better understand and manage their emotions. By providing a safe, judgment-free environment, EmoSense empowers users to express themselves freely and receive constructive emotional support at any time.

One of the standout features of EmoSense is its personalized chat history management, which allows users to title and revisit their emotional sessions, track progress over time, and gain deeper insights into their emotional journeys. The platform also includes profile customization, secure login, and session management tools to ensure a seamless and private user experience. With a focus on accessibility, privacy, and emotional intelligence, EmoSense leverages technology to create a reliable companion for anyone seeking comfort, clarity, and personal growth.

In addition to its core functionality, EmoSense incorporates role-based access for academic or administrative users—such as FYP Committee members—allowing them to monitor specific user interactions when authorized. This feature makes the platform suitable not only for personal use but also for academic research and evaluation. The system is developed using modern web technologies, integrated with Firebase for secure authentication and storage, and powered by Google’s Gemini API to provide emotionally intelligent chatbot responses.

Unlike many conventional mental health tools, EmoSense emphasizes continuous, real-time emotional support without requiring scheduled appointments or formal intervention. It is designed for everyday use, encouraging users to log their emotional states, reflect on their well-being, and develop greater emotional awareness over time. This level of accessibility fosters self-guided growth and resilience.

Ultimately, EmoSense bridges the gap between emotional need and digital convenience. By combining the scalability of AI with the sensitivity of human-centered design, EmoSense presents a new paradigm in emotional support—one that is proactive, personalized, and available to anyone, anywhere, at any time.

## 1.1 Brief

EmoSense is a web-based emotional support platform that utilizes artificial intelligence to assist users in understanding and managing their emotional well-being. Designed to function as a virtual emotional companion, EmoSense offers an intelligent chatbot interface that allows users to engage in private, empathetic conversations about their thoughts and feelings. This interaction is intended to simulate a supportive dialogue, helping users to express themselves freely and receive constructive, AI-generated feedback in return.

At the heart of the system is the integration of Gemini 1.5 Pro, a cutting-edge large language model developed by Google, which enables EmoSense to interpret user prompts and generate contextually relevant, emotionally intelligent responses. Unlike static support tools or generic chatbots, EmoSense offers dynamic and personalized engagement tailored to the emotional tone of the conversation.

The platform includes a number of supportive features such as user registration and login, profile customization (e.g., updating username and profile picture), and session management to provide a secure and user-focused experience. One of EmoSense's most distinctive capabilities is its chat history management system, which allows users to title and store past conversations, enabling them to monitor their emotional development over time. This makes the tool useful not only in the moment of interaction but also for long-term self-reflection and emotional tracking.

EmoSense has also been designed with academic integration in mind. It supports role-based access control, allowing members of an FYP (Final Year Project) Committee or academic supervisors to access specific student interaction data for evaluation purposes, while maintaining strict privacy controls.

The backend architecture of EmoSense is powered by MySQL, a robust and scalable relational database management system. All user information, chat records, roles, and session data are stored securely in structured tables, allowing for efficient retrieval, indexing, and relational querying. MySQL's support for complex joins and data integrity ensures that chat histories, profile information, and user roles remain consistent and protected. This structure also simplifies the generation of reports, user logs, and administrative controls, especially in multi-role academic environments.

The user interface adopts a clean, accessible, and responsive design, ensuring that users of all backgrounds can interact with the platform effortlessly. Whether it’s a student dealing with academic stress or an individual experiencing emotional burnout, EmoSense offers a reliable and judgment-free space to receive immediate, AI-powered emotional support.

By combining natural language processing, empathetic design, and a stable MySQL backend, EmoSense bridges the gap between emotional need and digital availability. It delivers a thoughtful, scalable, and user-centered solution to emotional well-being that is available anytime, anywhere.

## 1.2 Project Deliverables

The goal of this project was to develop EmoSense, a web-based emotional support platform that empowers users to manage, reflect on, and enhance their mental well-being through AI-assisted conversations. This platform was conceptualized to address the growing demand for accessible, stigma-free emotional support that is available anytime and anywhere. The final solution incorporates a range of functional modules, intelligent backend systems, and user-centered features, all designed to work in harmony to provide meaningful, secure, and real-time emotional assistance.

Key deliverables of the EmoSense project include:

- **Core Functional Modules:**
  The application includes several key modules, such as:
- **User Registration and Login**: Secure entry point for users using email and password authentication.
- **Profile Management:** Users can edit their profile details, including usernames and profile pictures.
- **AI-Powered Chatbot Interaction:** Real-time emotional conversations powered by the Gemini 1.5 Pro API, providing empathetic and context-aware responses.
- **Chat History Management:** Chats are saved with user-defined titles, allowing for easy reference and emotional progress tracking.
- **Role-Based Access Control:** Special access for FYP Committee members to review student interactions when necessary, supporting academic integration and evaluation.
- **Database and Backend Architecture:**
  The system’s backend is built using MySQL, which serves as the primary database for storing user data, chat history, session logs, and role definitions. The relational nature of MySQL allows for secure data integrity, efficient query performance, and scalable storage of structured emotional data.

- **System Documentation:**
- A **Software Requirements Specification (SRS)** document was created to clearly define all functional and non-functional requirements, covering user roles, security expectations, usability standards, and response behavior.
- A **Software Design Description (SDD)** was prepared to detail the system's architecture, data models, user flows, and interface design strategies. This includes Entity-Relationship Diagrams (ERDs), sequence diagrams, and navigation structures to guide implementation.
- **Frontend Interface:**
  The web-based UI is developed with a focus on accessibility, responsiveness, and simplicity. It allows users to seamlessly interact with the chatbot, manage their profile, and navigate between sessions without friction.
- **AI Integration:**
  The chatbot feature integrates Gemini 1.5 Pro to interpret user inputs, detect emotional tone, and respond in a manner that is both supportive and meaningful. This ensures that user interactions feel human-like and emotionally intelligent.
- **Testing and Evaluation:**
- **Unit Testing:** Validates individual components such as login validation, chat submission, and profile updates.
- **Integration Testing:** Ensures smooth interaction between modules, such as login-to-chat transitions and chat history storage.
- **Functional Testing:** Confirms that all features work as intended according to the requirement specifications.
- **Automated Testing:** Tools like Selenium and Postman were used to simulate and validate front-end behavior and API endpoints, respectively.
- **Security and Session Management:**
  EmoSense ensures secure handling of user sessions, password storage (with hashing techniques), and role-based access using session management logic implemented via backend technologies.
- **Final Working Application:**
  The completed platform provides users with a secure, private, and emotionally supportive environment. It empowers them to explore their emotions, monitor their psychological trends, and receive personalized guidance without judgment or delay.

This collection of deliverables reflects a holistic approach to emotional support software development, combining empathetic user experience design with robust technical architecture. EmoSense not only meets its intended goals but also lays the groundwork for future enhancements such as multilingual support, advanced analytics, and AI personalization.

## 1.3 System Limitations

While EmoSense offers a valuable and accessible tool for emotional support, like all systems, it has certain limitations that must be acknowledged for transparency, user awareness, and ongoing improvement.

- **Not a Replacement for Professional Therapy:**
  EmoSense is designed to provide emotional guidance through AI-driven conversation. However, it is not a certified mental health solution and should not be used as a replacement for therapy, counseling, or medical treatment. Users with severe mental health conditions are encouraged to seek professional help.
- **Dependence on Internet Connectivity:**
  The system operates entirely online and requires a stable internet connection. Users in remote areas or regions with poor connectivity may experience disruptions or an inability to access the platform.
- **Device Compatibility Requirements:**
  As a web-based application, EmoSense requires modern devices and updated browsers. Users with outdated smartphones, browsers, or low-end hardware may face difficulties in accessing or running the system smoothly.
- **AI Limitations and Bias:**
  Although the Gemini 1.5 Pro model is advanced, the chatbot's responses are based on machine learning models trained on existing data. As such, its understanding of user emotions may not always be accurate or culturally sensitive, and it may occasionally provide generalized or inappropriate responses.
- **Variation in User Experience:**
  The chatbot's effectiveness depends on the user's language, tone, and how they express emotions. Some users may find the responses helpful and engaging, while others may feel disconnected due to their communication style or expectations.
- **Legal and Ethical Boundaries:**
  All interactions and data collection must comply with relevant legal standards such as data protection laws (e.g., GDPR, HIPAA if applicable). Users should be made aware

of terms and conditions through clear consent forms and privacy policies.

- **Maintenance and Updates:**
  To ensure relevance and accuracy, the system requires regular updates and bug fixes. This includes maintaining chatbot performance, security patches, and improving user interface components. Failure to keep the system updated may reduce reliability over time.
- **Lack of Human Emotion Interpretation:**
  Unlike human therapists, the AI cannot pick up on nuanced non-verbal cues, sarcasm, or implicit emotions. This can result in limitations when trying to interpret subtle or ambiguous inputs.
- **Limited Language Support:**
  Currently, the system may only support English or a limited number of languages, potentially restricting accessibility for non-English speakers or multilingual users.
- **Scalability and Load Handling:**
  In the event of a large number of concurrent users, performance may degrade if the server infrastructure and database (MySQL) are not properly scaled. High server load may lead to slow response times or application crashes.

These limitations should be clearly communicated to users and stakeholders to ensure responsible use and realistic expectations. Recognizing these challenges also opens up opportunities for future development, such as multilingual support, offline capabilities, or AI model refinement.

## 1.4 Tools and Technology

These are all the tools and technologies which we're going to use throughout the project.

Example:

To develop the EmoSense platform, a wide range of modern tools and technologies were utilized, each selected to support different aspects of the system's design, development, and deployment. As shown in **Table 1.1**, tools like Figma were used to design UI mockups and wireframes, providing a clear layout of the system before coding began. Adobe Photoshop CS6 assisted in creating visual assets such as icons and logos that aligned with the platform's branding. For documentation and academic reporting, MS Word 2024 was used to draft detailed reports, while MS PowerPoint 2024 supported the preparation of project presentations. On the development side, HTML5, Bootstrap 5, and JavaScript were employed to build a responsive and interactive frontend. The backend logic was developed using PHP within the Laravel framework, which provided efficient routing, middleware, and session management. MySQL was integrated to handle secure database operations for user profiles, chat histories, and login details. At the core of the AI interaction lies the Gemini 1.5 Pro API, enabling the chatbot to deliver empathetic and context-aware responses. Together, these technologies ensured that EmoSense is user-friendly, intelligent, secure, and scalable—meeting the functional and emotional goals of the system.

Table 1.1 Tools and Technology

| | **Tools** | **Version** | **Rationale** |
|---|---|---|---|
| **Tools And Technologies** | Figma | Latest | Mock ups |
| | Adobe Photoshop | CSC 6 | Design Work |
| | MS Word | 2024 | Documentation |
| | MS Power Point | 2024 | Presentation |
| | API | 1.5 pro | Gemini |
| | HTML | 5 | Programming language |
| | Bootstrap | 5 | Platform |
| | JavaScript | | Query Language |
| | SQL | | Admin |
| | PHP | | Backend language |
| | Laravel | | Framework |

## 1.5 Relevance to Course Modules

The development of EmoSense brings together multiple domains within the Computer Science curriculum, offering a practical demonstration of how theoretical concepts are applied in real-world applications. It not only reinforces technical skills but also integrates ethical, design, and analytical aspects of modern computing education. Below is a detailed breakdown of its relevance across various modules:

- **Web Development**

  EmoSense demonstrates end-to-end web application development. Students applied skills in HTML5, CSS3, and JavaScript for the front end and Laravel (PHP framework) for the back end. Advanced routing, RESTful API handling, and UI responsiveness were applied to deliver a dynamic and user-friendly experience. This reinforces course learning in server-client architecture, form validation, and session-based navigation.

- **Artificial Intelligence**

  The use of Gemini 1.5 Pro, an advanced large language model, brings AI concepts to life. Topics like natural language understanding (NLU), contextual dialogue systems, and machine learning integration are applied. This helps students understand the practical implementation of NLP-driven AI agents in real-world emotional contexts, providing insight into chatbot design and intelligent interfaces.

- **Software Engineering**

  The EmoSense project follows formal SDLC models (e.g., Agile/Incremental), incorporating requirement gathering, analysis, modular design, implementation, testing, and deployment. Deliverables such as SRS, SDD, and testing reports align with the structured software engineering process taught in project-based modules, emphasizing scalability, maintainability, and traceability.

- **Human-Computer Interaction (HCI)**

  The project explores HCI principles through the development of an intuitive and emotion-sensitive chatbot interface. Topics like user empathy, user feedback loops, accessible design, minimal cognitive load, and emotional interface design are central to the UX choices made during development. The chatbot tone, feedback messages, and layout were intentionally crafted to reduce user anxiety and increase engagement.

- **Database Management Systems (DBMS)**

  EmoSense uses MySQL for data persistence. ER diagrams, foreign key constraints, normalization techniques, CRUD operations, and relational queries were implemented to manage users, chat logs, roles, and sessions. This reinforces concepts from database theory and lab practices.

- **Cybersecurity and Privacy**

  The system implements secure login, password encryption, session expiration, and role-based access control. Students gain hands-on experience applying security principles, user authorization flows, and GDPR-like consent mechanisms in line with cybersecurity module teachings.

- **Cloud Computing**

  While not primarily hosted on cloud, EmoSense can be extended to integrate cloud databases, AI endpoints, and deployment on cloud platforms like Heroku or AWS, linking to advanced modules like distributed computing and cloud-based web services.

- **Data Structures and Algorithms**

  Backend operations such as chat history sorting, session handling, and form validations rely on fundamental data structures (arrays, hash maps) and algorithmic logic (searching, filtering), as introduced in core DS&A courses.

- **Professional Practice and Ethics**

  EmoSense addresses real-world ethical concerns such as user consent, mental health sensitivity, bias in AI, and data handling responsibilities. These align with professional computing practices and the ethical decision-making process emphasized in curriculum modules on social and legal aspects of computing.

- **Capstone/FYP Project Management**

  As a final-year project, EmoSense involved milestone tracking, documentation, supervisor reviews, and formal evaluations. Students demonstrated the ability to translate academic knowledge into a professional-quality product while managing resources, timelines, and collaboration.

## 1.6 Project Background

With the global rise in mental health challenges such as anxiety, depression, and emotional burnout, access to immediate and reliable emotional support remains a significant barrier—particularly for students, young professionals, and individuals living in underserved regions. Traditional mental health services like therapy and counseling, while highly effective, are often expensive, stigmatized, and logistically inaccessible to a large portion of the population. As a result, many individuals are left to manage emotional distress alone, without tools or guidance to help them cope in real time.

EmoSense was conceived as a response to this growing need for accessible, stigma-free emotional assistance. By leveraging the principles of Affective Computing—a field introduced by Rosalind Picard that focuses on systems capable of recognizing and responding to human emotions—EmoSense integrates Emotional AI into a conversational interface that simulates empathetic dialogue. It is designed not to replace professional therapy but to serve as a digital companion that helps users process their feelings, reflect on their emotional states, and receive supportive feedback in a private and non-judgmental environment.

The project draws inspiration from leading emotional wellness tools like Wysa and Woebot, which have shown the power of chatbot-based interaction in reducing emotional stress and promoting self-awareness. EmoSense builds upon this idea by offering customizable features such as chat history tracking, personalized user profiles, and academic integration for research and feedback purposes. It also emphasizes emotional safety, ensuring that conversations are handled with sensitivity, neutrality, and privacy.

In line with current advances in conversational AI, EmoSense integrates Google's Gemini 1.5 Pro API to deliver contextually rich and emotionally aware responses. This enhances the user experience by providing not just scripted replies but dynamic, relevant support that adapts to the user's tone and input. With a web-based interface backed by a MySQL database, EmoSense is both scalable and secure, ensuring efficient data handling while maintaining a lightweight, responsive experience for the end user.

In summary, the development of EmoSense is grounded in both technological innovation and a social mission—to democratize emotional support through smart, ethical, and accessible AI solutions. By bridging the gap between digital convenience and emotional intelligence, the project aims to contribute meaningfully to the evolving conversation around mental health in the digital age.

## 1.7 Literature Review

Numerous studies and industry implementations highlight the growing relevance and effectiveness of Artificial Intelligence (AI) in the domain of mental health support. The intersection of Affective Computing, Natural Language Processing (NLP), and Conversational Agents has paved the way for the development of intelligent systems capable of offering emotional assistance, self-reflection tools, and guided mental wellness interventions.

Prominent AI-powered platforms such as Wysa, Woebot, and Replika have demonstrated that users are increasingly open to sharing their emotions with chatbots, particularly due to the judgment-free, private, and on-demand nature of these interactions. These tools often integrate Cognitive Behavioral Therapy (CBT) techniques and evidence-based emotional support strategies into automated dialogue flows, enabling them to provide not just emotional validation, but also structured coping mechanisms.

Studies such as those by Fitzpatrick et al. (2017) and Inkster et al. (2018) have shown that AI-based conversational agents can lead to measurable reductions in depression and anxiety symptoms when used consistently over time. These platforms rely heavily on NLP models to understand user inputs and respond with therapeutic scripts or interventions. However, many of these systems still depend on predefined flows or decision trees, limiting the depth and dynamism of conversation.

EmoSense builds upon this foundational work by introducing enhancements in the areas of contextual intelligence, user personalization, and data tracking. The platform leverages Gemini 1.5 Pro, a state-of-the-art large language model by Google, to generate responses that are not only grammatically and semantically accurate but also emotionally sensitive and context-aware. Unlike rule-based bots, Gemini enables EmoSense to carry on more fluid and meaningful conversations based on nuanced emotional cues.

Additionally, EmoSense addresses gaps in existing tools by focusing on:

- **Chat History Management:**

  While many platforms prioritize real-time interaction, EmoSense allows users to title, save, and review past conversations, promoting long-term emotional tracking and self-reflection—a feature particularly valuable in educational or research settings.

- **User Profile Customization:**

  EmoSense supports editable profile details such as usernames and profile pictures, allowing for a more personalized and user-centric experience. This enhances emotional connection and usability, especially for repeat users.

- **Academic Integration:**

  Unlike most commercial emotional AI tools, EmoSense includes role-based access for FYP Committee members, enabling faculty to evaluate user progress for academic research or mental wellness monitoring, all while respecting privacy boundaries.

Furthermore, EmoSense is designed with scalability in mind, utilizing a MySQL backend for structured data storage, and is built using modern web technologies to ensure cross-platform compatibility. Its architecture allows for future expansion, including potential integration of multilingual support, voice-based interfaces, or mood-tracking analytics.

In conclusion, the literature affirms the growing role of AI in digital mental health solutions. EmoSense contributes to this evolving field by enhancing contextual responsiveness, empowering user agency through personalization, and introducing academic utility—making it a uniquely positioned platform in the landscape of emotional AI.

## 1.8 Analysis from Literature Review

The literature clearly demonstrates that user trust, empathy, and engagement are essential elements in the effectiveness of AI-driven mental health tools. Studies on platforms like Wysa and Woebot highlight that users are more likely to return to and benefit from conversational AI tools when they feel understood, heard, and in control of their data. These findings have heavily influenced the design and functionality of EmoSense.

One key insight is that personalization significantly improves user comfort and emotional connection with digital agents. In line with this, EmoSense allows users to edit their profiles, including name and profile picture, helping to create a familiar, user-centric experience that supports long-term emotional journaling. This level of identity persistence gives users a sense of ownership and control, both of which are essential for engagement and trust in emotionally supportive systems.

Furthermore, the ability to title and manage chat histories is informed by the observed value of self-reflection and progress tracking in mental health management. While existing systems often emphasize real-time interactions, they lack mechanisms for users to revisit or contextualize past conversations. EmoSense fills this gap by letting users label their chat sessions—e.g., "Exam Stress" or "Feeling Lonely"—so they can return later to evaluate patterns in their emotional state over time. This aligns with therapeutic journaling strategies often used in counseling and CBT.

Another major takeaway from literature is the importance of emotional intent recognition—the ability of a system to infer not just what the user is saying, but how they feel. Research shows that empathetic response generation enhances user satisfaction and perceived effectiveness. EmoSense leverages the Gemini 1.5 Pro API for this purpose, enabling the chatbot to pick up on subtle cues in user input and respond with contextually appropriate emotional tone. This improves the perceived empathy of the system, which is strongly correlated with user retention and the system's supportive impact.

Security and privacy, as discussed in multiple academic papers, also play a major role in determining the credibility of digital health platforms. EmoSense addresses this by implementing secure authentication and structured database storage (via MySQL), along with clearly defined user roles (e.g., student vs. FYP Committee) to protect user data while supporting supervised academic use when necessary.

Additionally, the literature highlights a significant digital divide where access to emotional AI tools is limited by language, device compatibility, and internet access. While EmoSense currently operates as a web-based application with a focus on English, these insights support future enhancements such as multilingual support, offline logging, and mobile responsiveness to improve accessibility across diverse user demographics.

In summary, EmoSense is shaped by research-backed strategies that enhance user autonomy, emotional intelligence, and data transparency. By aligning system design with findings from successful implementations and academic literature, EmoSense positions itself as a reliable, ethical, and user-friendly platform for digital emotional support.

## 1.9 Methodology and Software Lifecycle for This Project

The development of EmoSense followed the Agile Software Development Methodology, which emphasizes flexibility, incremental progress, and collaborative feedback. This approach was chosen to ensure that the project could evolve responsively based on user needs and emotional design considerations—especially important in a domain as sensitive and human-centric as mental wellness.

Agile was particularly suited to EmoSense because it allowed the development team to focus on building and refining emotionally intelligent features over time, without being locked into a rigid waterfall structure. The methodology fostered a user-centered design philosophy, where real feedback from end-users and academic mentors was integrated at multiple points during development.

**Key Agile Principles Applied in EmoSense:**

- **Iterative Development**

  EmoSense was developed over a series of short, goal-driven sprints. Each sprint focused on delivering or improving specific modules, such as:

- Sprint 1: User authentication and login functionality.
- Sprint 2: Chat interface and chatbot integration using Gemini API.
- Sprint 3: Chat history titling and retrieval features.
- Sprint 4: Profile editing and session management.
- Sprint 5: FYP committee login and dashboard access.

These iterations allowed the team to test, receive feedback, and refine features with minimal delay.

- **Incremental Delivery**

  Rather than building the entire application before release, functionality was delivered in manageable increments. For example, the initial chatbot was integrated with basic response capabilities, then enhanced in later sprints with emotionally adaptive responses and smarter prompt analysis. This allowed for quicker deployment of a Minimum Viable Product (MVP) and continuous expansion based on testing outcomes.

- **Continuous Feedback**

  At the end of each sprint, feedback sessions were conducted with:

- **Student testers** (target users),
- **FYP Committee members** (evaluators)
- **Peers and instructors** (technical reviewers).

This input directly influenced enhancements such as the chatbot's response tone, the design of chat history titling, profile UI layout, and clarity of navigation. For example, feedback revealed that users appreciated a calming UI design, leading to UI adjustments for better readability and emotional comfort.

- **Adaptive Planning**

  The project roadmap was treated as a living document. As technical challenges arose (e.g., response time from API integration, MySQL query optimization), or as user expectations shifted (such as the need for clearer chat timestamps), tasks were reprioritized dynamically. This adaptive mindset ensured that the project continuously evolved to better meet its goals and users' emotional needs.

- **Cross-functional Collaboration**

  Agile promotes teamwork across design, development, and testing disciplines. EmoSense’s development involved collaboration between:

- Frontend developers (handling UI/UX),
- Backend developers (managing MySQL database and logic),
- AI integration handlers (for Gemini API),
- QA/testers (for sprint-based testing reports),

  ensuring a synchronized and efficient workflow.

- **Test-Driven Adjustments**

  With each sprint, testing (manual and automated) was integrated into the lifecycle to validate system stability and user experience. For instance, once the chatbot module was delivered, unit and integration tests were written and executed before moving to the next phase. Issues were logged, fixed, and reviewed within the same sprint.

The **Figure 1.1 illustrates** the Agile Software Development Lifecycle, a continuous and iterative approach used in the development of EmoSense. Agile methodology emphasizes collaboration, flexibility, and rapid delivery through short, focused sprints.

Each phase of the cycle—Plan, Design, Develop, Test, Release, and Feedback—was applied throughout the EmoSense project to support fast development while incorporating ongoing feedback from users and academic supervisors. This model allowed for regular updates, user-driven improvements, and effective integration of emotionally intelligent features such as chatbot communication, chat history storage, and user profile management.

Agile’s adaptability made it ideal for developing a system like EmoSense, where emotional tone, usability, and interface design benefit greatly from iterative testing and refinement.

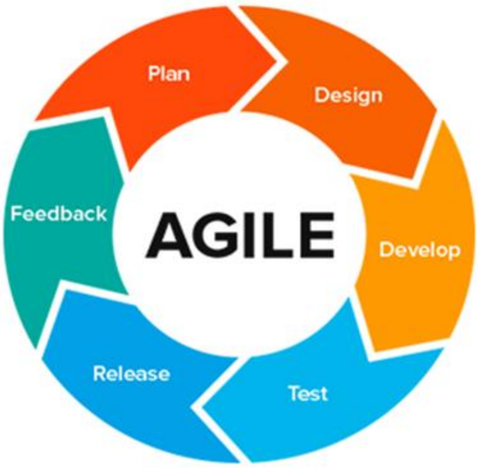


Figure 1.1 Agile Model Methodology

### 1.9.1 Rationale behind Selected Methodology

The Agile methodology was selected as the most appropriate development approach for the EmoSense project due to its flexibility, iterative progress, and alignment with the evolving nature of user needs in emotional wellness systems. The following points highlight the core reasons behind this decision:

- **Frequent Iterations for Continuous Improvement**

  Emotional AI applications, especially those involving chatbot interactions, require frequent refinements to tone, accuracy, and empathy. Agile's iterative structure allows for incremental updates to chatbot behavior after each sprint, making it easier to adjust responses, integrate feedback, and improve emotional sensitivity over time.

- **User-Centered Development**

  EmoSense is built for users seeking a safe, responsive, and empathetic emotional support tool. Agile prioritizes user stories and real-world feedback, which is ideal for a system that must continually adapt to meet the emotional and psychological expectations of its users. Feedback loops after each sprint ensured that features were designed not only for functionality but for comfort, accessibility, and trust.

- **Modular Architecture and Rapid Integration**

  The platform includes several independent components—login, chat interface, history management, and profile editing—all of which benefited from Agile's ability to develop and test features in modular sprints. Agile made it easier to integrate AI responses, refactor database connections (MySQL), and implement frontend changes without disrupting the entire system.

- **Cross-Functional Team Collaboration**

Agile’s structure supports collaboration between developers, testers, designers, and AI engineers. EmoSense involved:

- Frontend developers working on UI/UX,
- Backend developers handling session logic and MySQL queries,
- AI specialists integrating Gemini API for intelligent responses.

Agile sprints enabled synchronization across these domains, encouraging knowledge sharing and faster problem resolution.

- **Early Delivery of MVP (Minimum Viable Product)**

  Agile methodology supported the early release of a basic yet functional version of EmoSense. This Minimum Viable Product was used for demonstration, initial testing, and gathering early feedback from students and supervisors, helping to guide future development based on real usage insights.

- **Flexibility in Managing Evolving Requirements**

  Since emotional technology is a rapidly evolving field and user expectations shift quickly, Agile enabled the team to revise the development plan dynamically. New features such as chat titling, role-based access for committee members, or error handling for emotional inputs were added mid-project without disrupting the workflow.

- **Risk Mitigation and Quality Assurance**

  Agile's emphasis on frequent testing, such as unit and integration tests during each sprint, helped identify and fix bugs early. This reduced technical debt and ensured that each feature met performance, security, and user experience standards.

By choosing Agile, the EmoSense team was able to balance technological innovation with emotional design, delivering a robust, user-friendly, and emotionally aware platform within an academic timeline.

# Chapter 2
# Problem Definition

# 2. Problem Definition

## 2.1 Problem Statement

In today's fast-paced and digitally-driven world, mental health challenges such as stress, anxiety, and emotional fatigue are becoming increasingly prevalent across all age groups. Despite growing awareness, access to professional mental health care remains limited for many individuals due to factors such as high costs, long waiting times, geographic inaccessibility, and most notably, social stigma. As a result, a significant number of people suffer in silence, lacking the tools or support needed to cope with emotional distress.

Furthermore, traditional mental health support systems often require scheduled appointments, in-person visits, or clinical environments, which may not be suitable or comfortable for everyone—particularly those who are hesitant to seek formal help. The absence of immediate, judgment-free support often leads to emotional suppression or unhealthy coping mechanisms.

There is a growing demand for digitally accessible, emotionally intelligent, and user-centric platforms that allow users to express their emotions freely, receive empathetic responses, and reflect on their mental states over time. While existing AI-powered mental health tools like Wysa and Woebot have made significant strides, there is still a lack of personalization, contextual awareness, and academic adaptability in many solutions.

### 2.1.1 Problem Solution

To address the growing gap in accessible, affordable, and stigma-free emotional support, the proposed system—EmoSense—offers a web-based platform that uses AI to simulate empathetic, emotionally aware conversations. The core idea is to provide a space where users can express their emotions freely, reflect on their experiences, and receive real-time support without the need for professional intervention or human judgment.

At the center of this system is a chatbot powered by Gemini 1.5 Pro, capable of engaging in context-sensitive and emotionally intelligent interactions. Users can describe their emotional state, talk about personal issues, or seek advice, and the chatbot responds in a supportive, conversational tone that mimics therapeutic dialogue. This enables users to feel heard and validated, which is often the first step toward emotional healing.

In addition to real-time conversation, EmoSense includes several supportive features designed to enhance emotional self-awareness and management:

- **Chat History with Titles:** Users can save conversations under custom titles (e.g., "Feeling Overwhelmed" or "Post-Exam Stress") to revisit later, fostering self-reflection and pattern recognition in emotional behavior.

- **Profile Customization:** A personalized interface allows users to edit their username and profile image, making the platform feel more relatable and familiar.
- **Role-Based Access:** The system supports academic usage, where designated FYP Committee members can securely view specific student interactions for evaluation or support, without breaching confidentiality.
- **Resource Integration:** Alongside chatbot guidance, the platform may suggest coping strategies such as breathing exercises, mindfulness prompts, or motivational techniques. It can also direct users to external professional resources when more serious intervention is needed.
- **Privacy and Anonymity:** Users interact with the system through secure login, ensuring a private and safe environment. Sensitive emotional data is securely stored using MySQL, with role-based access and encrypted session handling.

## 2.2 Objectives of the Proposed System

The EmoSense platform is designed with the primary goal of enhancing emotional wellness through AI-assisted interactions in a safe, private, and accessible digital space. The following behavioral objectives (BO) outline the key functions, goals, and values that the system aims to fulfill:

- **BO-1: Provide AI-Guided Emotional Support via Chatbot**
  Leverage the Gemini 1.5 Pro API to simulate human-like, empathetic conversations, allowing users to freely express emotions and receive emotionally intelligent guidance anytime, anywhere.
- **BO-2: Deliver Personalized, Emotionally Relevant Responses**
  Use natural language understanding and emotional context analysis to tailor responses to each user's emotional state, fostering meaningful engagement and promoting emotional regulation.
- **BO-3: Enable Chat History Saving and Tracking**
  Allow users to title, save, and revisit past conversations, supporting long-term reflection, emotional pattern recognition, and personal growth.
- **BO-4: Offer Profile Personalization**
  Empower users to customize their profile (e.g., username and profile picture) to create a familiar, user-centric environment that enhances their comfort and ongoing engagement with the platform.

- **BO-5: Ensure Security and Confidentiality of Emotional Data**

  Use secure login mechanisms and structured MySQL-based data storage to protect sensitive user data and ensure that all emotional disclosures remain private and encrypted.
- **BO-6: Promote Emotional Self-Care through Continuous Availability**

  Provide an always-on, responsive space where users can access emotional support at their convenience, reducing the feeling of isolation and improving mental health consistency.
- **BO-7: Support Academic Integration with Role-Based Access Control**

  Allow designated academic users (e.g., FYP Committee members) to securely view specific student interaction data for evaluation purposes, supporting mental health monitoring in educational settings.
- **BO-8: Encourage Consistent Emotional Check-Ins**

  Motivate users to return regularly to log their feelings, track changes in mood, and develop emotional awareness through guided interaction and progress visualization.
- **BO-9: Facilitate Safe and Non-Judgmental Conversations**

  Ensure that the chatbot maintains a neutral, empathetic, and supportive tone, making users feel heard, accepted, and encouraged to open up without fear of judgment.
- **BO-10: Lay Foundation for Future Enhancements**

  Design the platform architecture in a modular, scalable way that allows for future upgrades such as multilingual support, voice interactions, mood analytics, and professional escalation features.

## 2.3 Scope

EmoSense is a web-based emotional support system designed to offer a secure, accessible, and intelligent platform for users seeking assistance with mental and emotional well-being. The scope of the project covers both core functional modules and supporting features, ensuring that users can express themselves, receive empathetic responses, and maintain emotional self-awareness in a private digital space.

- **Functional Scope**
- **Emotion Recognition and Context Interpretation**

  EmoSense integrates AI models capable of detecting emotional tone from user inputs, allowing for a more personalized and empathetic response. The chatbot analyzes language patterns to infer emotions such as sadness, anxiety, or frustration, tailoring its support accordingly.

- **AI-Driven ChatBot Interaction**

  Powered by Gemini 1.5 Pro, the chatbot conducts real-time, human-like conversations to guide users through emotional expression and self-reflection. The chatbot can respond to a wide variety of prompts with contextual relevance, ensuring that users feel heard and validated.

- **Chat History Management**

  Users can save conversations, assign titles (e.g., "Exam Stress"), and revisit previous sessions to track their emotional journey. This feature supports journaling-style reflection and encourages consistent engagement with one's mental health.

- **User Profile and Personalization**

  EmoSense allows users to edit their usernames and upload profile pictures, enhancing personalization and familiarity. Profiles help maintain continuity across sessions while giving users a sense of ownership over their emotional space.

- **User Authentication and Session Management**

  The system includes a secure login and logout mechanism, ensuring that only authorized users access sensitive content. Session persistence and proper role-based access (e.g., Student, FYP Committee) are implemented to protect data and control permissions.

- **Resource Library (Optional/Scalable)**

  A module for curated resources—such as articles, videos, mindfulness exercises, and self-care tips—is included to supplement chatbot support with educational content, making EmoSense not only a conversational tool but also a learning platform for mental health.

- **Admin/FYP Committee Access (Role-Based)**

  Academic evaluators can log in under specific roles to access student interaction histories (with appropriate permissions), facilitating evaluation or mental health monitoring as part of FYP requirements.

- **Technical and Non-Functional Scope**
- **Database Management with MySQL**

  EmoSense uses MySQL for structured, scalable, and secure data management. It stores user profiles, chat sessions, emotional tags, and role-based permissions in a normalized schema optimized for performance and integrity.

- **Web Responsiveness and Accessibility**

The platform is designed to be fully responsive, adapting to various devices (desktops, tablets, smartphones) and screen sizes. This ensures ease of access regardless of the user's device or physical location.

- **Data Privacy and Security**

  All user interactions are encrypted and securely stored, and the application enforces strict session handling to protect against unauthorized access. Users are made aware of privacy policies and terms of service before interacting with the platform.

- **Scalability and Extendibility**

  The modular architecture allows future enhancements such as:

- Multilingual support
- Voice-based chatbot interaction
- Mood trend analytics
- Mobile app deployment
- Integration with professional help (referrals)

**Out of Scope**

- **Diagnosis or Treatment of Clinical Conditions**

  EmoSense does not provide formal mental health diagnoses or replace professional therapy. It is meant for emotional guidance and self-help, not clinical intervention.

- **Offline Access**

  As a web-based system, EmoSense requires internet connectivity to function and does not currently support offline use.

By defining a clear and focused scope, EmoSense ensures delivery of a robust, user-centered, and ethically sound emotional support platform, suitable for both individual users and academic research purposes.

## 2.4 Deliverables and Development Requirements

### 2.4.1 Deliverables:

The EmoSense project aims to deliver a comprehensive, web-based emotional support platform that integrates intelligent features with user-centric design. The following are the key deliverables achieved throughout the development lifecycle:

- **Fully Functional Web Platform with User Authentication**

A responsive and secure web-based application has been developed, enabling users to:

- **Register** for new accounts
- **Log in/log out** using valid credentials

- **Maintain secure sessions** with appropriate timeouts and session persistence
  The authentication system also supports role-based access control, distinguishing between regular users (students) and academic evaluators (FYP Committee).
- **AI Chatbot Integrated with Gemini 1.5 Pro**

The platform incorporates Google's Gemini 1.5 Pro API, enabling:

- Real-time, emotionally aware conversations
- Contextual responses based on the tone and content of user inputs
- Dynamic, intelligent interactions that simulate therapeutic dialogue
  This integration ensures that the chatbot can adapt to varying emotional states and deliver supportive, human-like communication.
- **Chat History Management with Titles**

Users can:

- Save their chat sessions
- Assign custom titles (e.g., "Feeling Low", "Exam Stress") to sessions
- Revisit previous conversations for self-reflection and emotional tracking
  This feature supports the emotional journaling aspect of the platform and encourages ongoing emotional engagement.
- **Profile Editing and Personalization Features**

To enhance user comfort and connection with the platform, EmoSense includes:

- Editable **usernames**
- Uploadable **profile pictures**
- Persistent profile data across sessions
  These features provide a more **personalized and user-friendly experience,** encouraging regular use and emotional continuity.
- **Role-Based Dashboard for FYP Committee**

The system provides academic evaluators with:

- Secure login access
- The ability to **view student chat** histories (when authorized)
- Insights into the emotional engagement and usage patterns of students
  This facilitates research, feedback, or monitoring in an ethical and structured way.

**MySQL Database Integration**

A structured **MySQL backend** stores:

- User data and credentials
- Chat logs and session metadata

- Role assignments and system settings

  The database is designed with **normalized schema**, optimized queries, and appropriate constraints to ensure data integrity, security, and scalability.

- **Testing and Documentation**

Complete documentation has been developed, including:

- Software Requirements Specification (SRS)
- Software Design Description (SDD)
- Test cases for unit, integration, functional, and automated testing
- Project report with architecture diagrams, use cases, and design models

  These deliverables ensure the maintainability, reproducibility, and extensibility of the system.

### 2.4.2 Development Requirements:

The development of EmoSense required the integration of various technologies and tools across frontend, backend, database, and AI services. Each component was carefully selected to ensure responsiveness, data security, intelligent interaction, and scalability. The following are the core requirements and tools used in the system:

- **Frontend Technologies**
- **HTML5**

  Used to structure the content and layout of all web pages, ensuring semantic markup and accessibility.

- **CSS3**

  Applied for responsive design and styling of components including forms, chatbot interface, buttons, and modals. Media queries were used to adapt the layout across devices.

- **JavaScript (Vanilla + DOM Manipulation)**

  Enabled client-side interactivity such as form validation, real-time message handling, and dynamic updates to user input fields.

- **Blade Templates (Laravel's templating engine)**

  Used for managing dynamic content rendering within the Laravel framework, promoting reusability and separation of concerns.

- **Backend Technologies**
- **Laravel (PHP Framework**)
- Provides MVC architecture for clean separation of logic, presentation, and data.

- Includes built-in authentication features such as login, registration, and password reset.
- Used for handling HTTP requests, routing, form submissions, and middleware logic.
- **MySQL (Relational Database)**
- Stores structured data including user accounts, chat logs, emotional tags, session data, and role permissions.
- Ensures data integrity through the use of foreign keys, indexes, and normalization.
- Supports relational queries used in user history retrieval and admin-level analytics.
- **AI Integration**
- **Gemini 1.5 Pro API (Google's LLM)**
- Powers the chatbot responsible for analyzing and responding to user prompts with emotionally intelligent responses.
- Offers natural language understanding (NLU) and generation (NLG), enhancing the empathy and contextual relevance of chatbot replies.
- Interacts with Laravel backend via secure API calls to process chat input and return dynamic responses.
- **Security & Session Management**
- **Laravel Auth Middleware**
- Ensures only authorized users can access certain routes.
- Supports role-based access control (e.g., student vs. FYP Committee).
- **CSRF Protection & Form Validation**
- Laravel's CSRF tokens protect all user-submitted forms.
- Backend and JavaScript validations safeguard against malformed or malicious inputs.
- **Session Handling**
- User sessions are stored securely and expire automatically after inactivity.
- User roles are preserved across navigation using session data.
- **Development Tools & Environments**
- **Visual Studio Code** – Primary IDE for frontend and Laravel backend development.
- **XAMPP / Laravel Valet / Artisan CLI** – For local server environment and Laravel routing.
- **Postman** – Used to test Gemini API responses and internal Laravel API endpoints.
- **Git & GitHub** – Version control and collaboration.
- **Google Chrome DevTools** – For real-time debugging and performance analysis.

These development requirements collectively enabled the creation of a responsive, intelligent, and secure emotional support platform. The chosen tech stack supports current functionality and leaves room for future upgrades, including real-time socket communication, multilingual chatbot support, and mobile application deployment.

## 2.5 Current System

The EmoSense system is a web-based emotional support platform developed to provide accessible, private, and intelligent mental health assistance through conversational AI. Drawing inspiration from widely recognized mental health chatbots like Wysa and Woebot, EmoSense aims to replicate their strengths while addressing some of their limitations. Wysa offers structured self-help using cognitive behavioral therapy (CBT) tools, and Woebot engages users with friendly, psychologically informed check-ins. EmoSense similarly allows users to express their emotions through natural language prompts and receive empathetic, context-aware responses powered by the Gemini AI model. For instance, when a user types "I feel anxious about exams," EmoSense responds with supportive guidance such as, "That's completely understandable—would you like to talk about what's making you feel this way?" This interaction helps users feel heard and supported without judgment.

What sets EmoSense apart is its academic orientation, offering features like custom-titled chat histories for emotional tracking, secure authentication and data storage via Firebase, and role-based access for FYP Committee members to review user interactions when required. Users can also customize their profiles, edit their usernames, and upload profile pictures to personalize their experience. In addition, EmoSense provides a safe digital environment for emotional self-expression, with strong privacy controls and user-focused features. Unlike some competitors, it encourages long-term reflection by allowing users to revisit past emotional states through saved conversations. The chatbot is not limited to predefined flows—it can engage in more dynamic and flexible conversations, offering a sense of companionship rather than scripted support. EmoSense is also designed to be extensible, with future plans to integrate multilingual capabilities and mobile compatibility. By combining the conversational intelligence of modern AI with secure data practices and user-centered design, EmoSense serves as both a compassionate companion and a meaningful academic tool for exploring emotional well-being. There is no existing system for EmoSense. However, platforms like Wysa and Woebot serve as indirect references. These platforms, while effective, often lack full customization features or open-ended emotional exploration like EmoSense intends to offer.

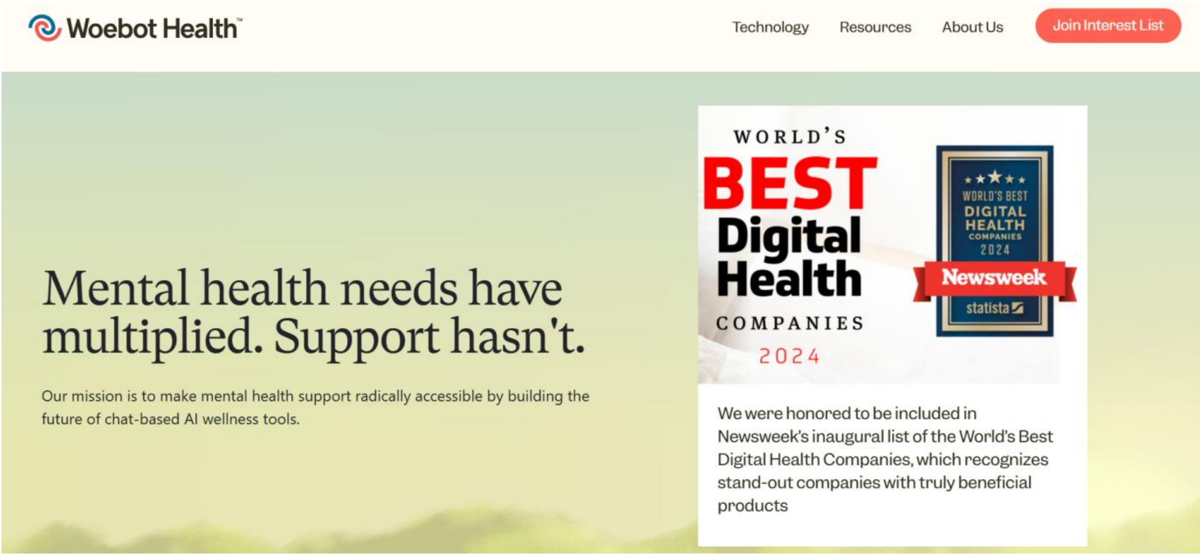


Figure 2.1 WoeBot Chatbot

# Chapter 3
# Requirement Analysis

# 3. Requirement Analysis

EmoSense requires core functionalities such as user registration/login, AI-driven chatbot interaction, emotional prompt processing, chat history management with titled sessions, profile editing, and secure session handling. Role-based access is also included for FYP Committee members to review user interactions when authorized. The Use-Case approach is used to capture system interactions, detailing how users engage with features like starting a chat, editing profiles, and reviewing past sessions. This method ensures all functional requirements are clearly defined, supporting a smooth and secure emotional support experience.

## 3.1 Use Cases Diagram

The Use Case Diagram provides a high-level visual representation of the interactions between the users and the EmoSense system. It identifies the primary actors (users, system admin) and the core functionalities of the platform. The use case diagram of EmoSense represents how different users interact with the system to perform key functions. The main actor is the User, who can register, log in, and access features such as emotion recognition, chatbot interaction, the resource library, and privacy settings. The user also undergoes authentication to ensure secure access. The Bot assists specifically in the chatbot interaction use case, helping users by processing and responding to emotional prompts. The Creator plays a role in managing user authentication and ensuring privacy within the system. This **Figure 3.1** provides a clear overview of how EmoSense enables emotional support, secure access, and personalized user interaction through collaborative system components.

Usecase Diagram Of EmoSense

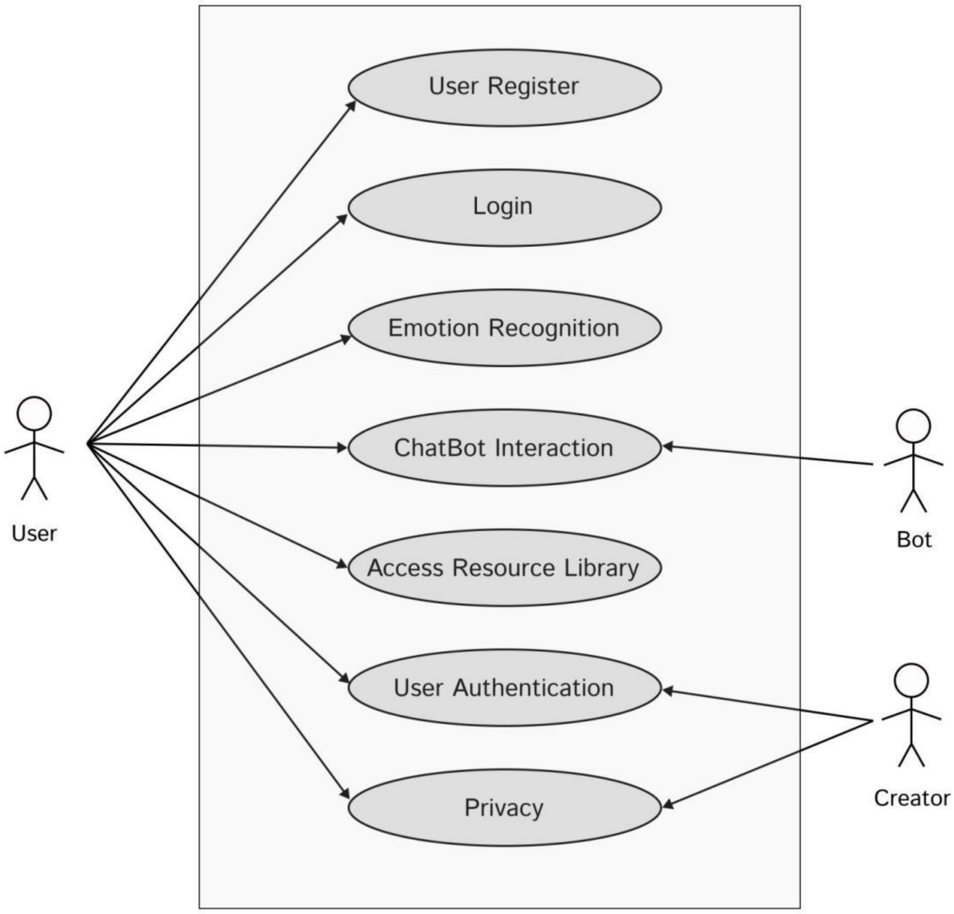


Figure 3.1 Use Case Diagram

## 3.2 Use Case Description

The table below indicates comprehensive use case descriptions.

**Table 3.1** describes the Registration use case for the EmoSense platform. It outlines the interaction where a user initiates the registration process, provides login credentials, and is validated by the system to gain access. The table includes the normal and alternative flows, exceptions like incorrect credentials, and emphasizes the requirement for a network connection.

Table 3.1 Login

| | |
|---|---|
| Use case id | 01 |
| Use case name | Registration |
| Actors | User |

| | |
|---|---|
| Description | A user registers on the EmoSense platform. |
| Trigger | User requests to register. |
| Pre-condition | The user has an account. |
| Postcondition | The actor is now logged into the system. |
| Normal flow | This case starts when an actor wants to login into the system.<br>The system requests that the actor enter e-mail/phone no. and password.<br>The actor enters-mail/phone number and password.<br>The system validates the entered e-mail/phone no. and password and<br>login into the system. |
| Alternative flow | For example, a Valid e-mail/Phone number or an Invalid Password If in the basic flow the system cannot find the valid e-mail or thepassword is invalid, an error message is displayed. |
| Exceptions | Incorrect credentials.<br>Request to re-enter the login details. |
| Business rules | The system must be connected to the network. |

**Table 3.2** outlines the Emotion Analysis use case within the EmoSense platform. It details how the system processes user-submitted messages to detect emotional states using the Emotion Analyzer Service. Once classified, the emotion is used to generate context-aware responses. The table includes normal and alternative flows, exceptions like service unavailability, and rules ensuring language support and emotion detection accuracy.

Table 3. 2 Emotion Analysis

| | |
|---|---|
| Use case id | 02 |
| Use case name | Emotion Analysis |
| Actors | User, Emotion Analyzer Service |
| Description | The system analyzes the user's input to detect emotional states. |
| Trigger | The user submits a query or message through the platform. |
| Pre-condition | The user is logged in and provides text input. |
| Postcondition | The emotion is identified and passed to the ChatBot module for response generation. |
| Normal flow | 1. User submits a message.<br>2. The system sends the message to the emotion analyzer.<br>3. The analyzer processes the input and classifies the emotion. |

| | |
|---|---|
| | 4. The classified emotion is logged and used for generating personalized responses. |
| Alternative flow | 1. Input is ambiguous.<br>2. The system requests clarification from the user. |
| Exceptions | 1. Emotion analyzer service is unavailable.<br>2. Input data is insufficient. |
| Business rules | 1. Input must be in a supported language.<br>2. System must classify emotions within a predefined confidence threshold. |

**Table 3.3** describes the ChatBot Interaction use case in the EmoSense system. It captures how users engage with the AI-powered ChatBot to receive personalized emotional support. The flow involves the user sending a query, the system processing it via the ChatBot module, and logging the response. It also outlines alternate scenarios such as misunderstood queries and exceptions like connection issues, while ensuring privacy and contextual relevance in all interactions.

Table 3. 3 Chatbot Interaction

| | |
|---|---|
| Use case id | 03 |
| Use case name | ChatBot Interaction |
| Actors | User, ChatBot Module |
| Description | The user interacts with the AI-powered ChatBot for personalized mental health support. |
| Trigger | User initiates a conversation with the ChatBot. |
| Pre-condition | The user is logged in and the ChatBot is online. |
| Postcondition | ChatBot provides a personalized response and logs the interaction. |
| Normal flow | 1. User asks a question.<br>2. The system sends the query to ChatBot module.<br>3. ChatBot generates a response.<br>4. The response is displayed to the user.<br>5. The interaction is logged in the database. |
| Alternative flow | 1. The ChatBot fails to understand the query.<br>2. The system requests the user to rephrase the query. |
| Exceptions | 1. ChatBot module is temporarily unavailable.<br>2. Users lose connection during the interaction. |
| Business rules | 1. Responses must be relevant and contextually accurate.<br>2. User privacy is maintained during interactions. |

**Table 3.4** outlines the Resource Recommendation use case in the EmoSense platform. It describes how the system, based on the user's detected emotional state from ChatBot interactions, suggests relevant mental health resources. The process involves querying a categorized resource library and displaying personalized content. It includes alternative flows for unmatched queries and exceptions like module unavailability, while adhering to rules

ensuring relevance and user preference consideration.

Table 3. 4 Resource Recommendation

| | |
|---|---|
| Use case id | 04 |
| Use case name | Resource Recommendation |
| Actors | User, Resource Library Module |
| Description | The system recommends mental health resources based on the user's emotional state. |
| Trigger | ChatBot identifies a need for additional resources. |
| Precondition | The user has interacted with ChatBot, and emotion data is available. |
| Postcondition | The user receives a list of relevant resources. |
| Normal flow | 1. ChatBot identifies the need for resources.<br>2. The system queries the resource library.<br>3. The library retrieves relevant resources based on emotion type.<br>4. The system displays the resources to the user. |
| Alternative flow | 1. No resources match the query.<br>2. The system suggests general resources. |
| Exceptions | 1. Resource library module is unavailable.<br>2. Database query fails. |
| Business rules | 1. Resources must be categorized and relevant.<br>2. User preferences are considered when recommending resources. |

**Table 3.5** describes the Professional Support Referral use case in the EmoSense system. It details how the platform assists users in connecting with mental health professionals when the chatbot identifies the need or the user explicitly requests help. The system presents a list of professionals or emergency contacts, depending on availability. It also handles exceptions like referral service downtime, while ensuring that all referrals comply with privacy regulations and require user consent

Table 3. 5 Professional Support Referral

| | |
|---|---|
| Use case id | 05 |
| Use case name | Professional Support Referral |
| Actors | User, Referral Module |
| Description | The system connects users to professional support if needed. |
| Trigger | User requests professional help or chatbot identifies the need. |
| Pre-condition | User is logged in and agrees to referral terms. |
| Postcondition | User receives professional contact details or is redirected to an external service. |

| | |
|---|---|
| Normal flow | 1. User requests support.<br>2. The system displays a list of available professionals.<br>3. User selects and contacts the professional. |
| Alternative flow | No professionals available; the system provides emergency contact information. |
| Exceptions | Referral service unavailable. |
| Business rules | Referrals must adhere to privacy laws and user consent requirements. |

## 3.3 Functional Requirements

These are the following modules that will be included in this system.

### 3.3.1 Emotion Recognition Module

This module analyzes user input to detect emotions using AI techniques.

- Analyze user text for emotional cues.
- Classify and report detected emotions.
- Provide feedback based on emotional analysis.

### 3.3.2 ChatBot Interaction Module

This module allows users to interact with a ChatBot that provides mental health support.

- Respond to user queries with relevant advice.
- Maintain conversation context to provide coherent support.
- Offer suggestions based on detected emotions.

### 3.3.3 Resource Library Module

This module provides users with access to mental health resources.

- Display a categorized list of resources
- Allow users to search for specific topics.
- Provide detailed information and links to external resources.

### 3.3.4 User Authentication and Privacy Module

This module ensures secure user authentication and protects user data.

- User registration and login.
- Encrypt user data and conversation logs.
- Implement privacy policies to protect user information.

## 3.4 Non-Functional Requirements

Non-Functional Requirements are the quality or abstract qualities that should be in your system. Non-functional requirements explain the constraints of the system. It identifies the attributes under which the system is operating. Some of the non-functional requirements for the auto shipping management system are given below.

### 3.4.1 Usability

Usability is a cornerstone of the EmoSense platform, as it directly impacts the user's comfort, emotional engagement, and likelihood of returning for support. EmoSense has been designed with a clean, intuitive, and minimalistic interface to ensure that users from all age groups and technical backgrounds can interact with the system effortlessly.

The interface uses a dark mode theme with soft contrast to reduce visual strain, especially for users accessing the platform during emotional distress or at night. Core functions such as starting a new chat, accessing saved conversations, editing profiles, and logging out are placed in easily accessible areas with clearly labeled icons and buttons.

Navigation is streamlined, with collapsible menus and a guided layout that walks users step-by-step through chatbot interaction. Tooltips, placeholders, and contextual prompts are integrated throughout the application to guide users without overwhelming them. EmoSense also includes keyboard accessibility, large clickable areas, and responsive design, ensuring that the platform works smoothly across desktops, tablets, and smartphones.

In addition, profile customization features, such as uploading a profile picture and updating usernames, make the experience more personal and emotionally grounding. These design decisions aim to build a sense of belonging and trust between the user and the platform.

Above all, EmoSense strives to create an inclusive and emotionally safe environment, where users can focus on their well-being without confusion or technical friction. Whether it's a student facing exam anxiety or a user seeking silent companionship, the platform ensures that support is just a few clicks away—clearly, comfortably, and reliably.

### 3.4.2 Performance

Performance is a critical factor for user satisfaction in any interactive web application—especially in platforms like EmoSense, where emotional support is expected to be immediate, consistent, and reliable. To maintain a responsive and seamless experience, EmoSense is designed to deliver system responses within an optimal time frame—typically under 2 seconds, even during periods of high user activity.

EmoSense achieves this performance benchmark through several key strategies:

- **Optimized Backend with Laravel and MySQL**

  The backend infrastructure uses Laravel, a powerful PHP framework known for its speed and modularity. Efficient MySQL queries are implemented with indexing and normalization to reduce data retrieval time and maintain low server load.

- **Fast API Interaction with Gemini 1.5 Pro**

  The integration with the Gemini 1.5 Pro API is designed for rapid, asynchronous request handling. Chat inputs are processed and returned using lightweight endpoints to ensure that AI-generated responses are delivered in real time, preserving the flow of conversation without delay.

- **Caching and Session Optimization**

  Frequently accessed elements, such as chat history previews and user profiles, are temporarily cached to reduce repeated database calls. Sessions are managed securely but efficiently, minimizing processing overhead for each logged-in user.

- **Responsive Frontend and Lightweight Components**

  EmoSense uses clean HTML5, Bootstrap, and JavaScript to ensure fast rendering on both desktop and mobile devices. Animations and transitions are kept minimal to avoid UI lag, while the use of AJAX in form submissions prevents full-page reloads.

- **Scalable Architecture**

  The system is designed with scalability in mind, allowing for vertical or horizontal scaling as user demand increases. This ensures that multiple users can interact with the chatbot simultaneously without affecting load time or responsiveness.

In emotionally sensitive applications like EmoSense, speed is not just a technical metric—it's an essential part of delivering comfort, reassurance, and emotional continuity. A lag in response can break the user's emotional flow, while a fast reply can help maintain engagement and trust. EmoSense's performance-driven design ensures that users receive timely, empathetic support whenever they need it, without interruption or frustration.

### 3.4.3 Security

Security is a foundational pillar of the EmoSense platform, especially considering the sensitive and emotionally personal nature of user data. Users must feel confident that their emotional disclosures, identity, and behavioral patterns are protected from misuse or exposure. To that end, EmoSense incorporates a multi-layered security framework designed to protect user data, prevent unauthorized access, and maintain trust throughout the user experience.

Key security features and practices implemented in EmoSense include:

- **End-to-End Data Encryption**

  All user data transmitted between the client and server is secured using HTTPS with

SSL/TLS protocols, ensuring that messages and credentials are encrypted and safe from interception during transit.

- **Secure Authentication and Session Management**
  The system uses Laravel's built-in authentication mechanism, supporting hashed password storage (via bcrypt) and secure session handling. Sessions are automatically timed out after inactivity to reduce risks of unauthorized access on shared devices.
- **Role-Based Access Control (RBAC)**
  EmoSense implements RBAC to differentiate access permissions for students, administrators, and FYP Committee members. This ensures that sensitive user interactions are only accessible to authorized roles, and academic evaluators can only view permitted content for evaluation purposes
- **Data Storage Security with MySQL**
  Sensitive data, such as chat logs and user profiles, are securely stored in MySQL databases using proper indexing, prepared statements, and parameterized queries to prevent SQL injection attacks.
- **Input Validation and Sanitization**
  All user inputs are validated and sanitized on both the client and server sides to prevent XSS (Cross-Site Scripting) and injection-based vulnerabilities.
- **Regular Security Audits and Code Reviews**
  EmoSense undergoes periodic security audits and vulnerability assessments, including code reviews and testing against OWASP Top 10 threats. These proactive steps help detect and patch any weaknesses before they can be exploited.
- **Privacy Policy and Consent Management**
  The application clearly communicates its privacy policy, informing users about data collection, storage, and usage practices. Before using sensitive features (like saving chat history or sharing with committee members), explicit user consent is obtained.
- **Error Handling and Logging**
  The system avoids exposing technical errors to users, preventing leakage of internal logic or structure. Errors are logged securely for administrative review while showing safe, user-friendly messages to the frontend.

By combining technical best practices with ethical data handling, EmoSense ensures a secure environment where users can express themselves without fear of exposure or misuse. Security is not treated as an afterthought—it is built into every layer of the platform, reinforcing EmoSense’s mission to be a trustworthy and emotionally safe companion for all users.

### 3.4.4 Portability

Portability is a key non-functional requirement for EmoSense, ensuring that users can access emotional support services without being restricted by device type, operating system, or location. The platform is developed as a responsive, browser-based web application, making it highly compatible with various environments such as Windows, macOS, Linux, Android, and iOS.

EmoSense is fully accessible through all major modern web browsers, including:

- Google Chrome
- Mozilla Firefox
- Safari
- Microsoft Edge

Its interface automatically adapts to various screen sizes, ensuring a seamless experience whether accessed from a desktop monitor, laptop, tablet, or smartphone. This responsiveness is made possible by the use of HTML5, CSS3, JavaScript, and Bootstrap 5, which work together to deliver a fluid layout and touch-friendly controls across platforms.

Bright Start: Whether users are seeking support from the comfort of their home on a laptop or while traveling with their mobile device, EmoSense provides a consistent and immersive user experience. There is no need for separate downloads or installations—simply visiting the web address provides full functionality.

The platform is also optimized for low-bandwidth and limited-resource environments, ensuring that even users on older devices or slower internet connections can still receive timely and reliable support. All visual elements and data requests are lightweight, reducing load times and conserving device resources.

In the future, EmoSense has the potential to be extended into a Progressive Web App (PWA) or deployed as a native mobile application using cross-platform frameworks such as Flutter or React Native, further enhancing accessibility and offline capabilities.

By embracing portability, EmoSense guarantees that emotional wellness support is truly universal and on-demand—meeting users wherever they are, whenever they need it, and on whatever device they feel most comfortable using.

### 3.4.5 Reliability

Reliability is critical for a mental health support platform like EmoSense, where users may depend on the system during vulnerable or emotionally urgent moments. EmoSense is designed to offer continuous, dependable service with minimal interruptions, ensuring that users can trust the platform to be available whenever support is needed.

To ensure high reliability, EmoSense is built on a robust backend infrastructure using Laravel and MySQL, which allows for secure and consistent handling of user data and chatbot interactions. The system architecture supports redundant services and error handling mechanisms, reducing the risk of system crashes or data loss during unexpected events.

Bright Start: EmoSense uses automated daily backups, ensuring that all user data—including chat histories, login information, and profile settings—is preserved and can be quickly restored in the event of a failure. This not only protects data integrity but also reinforces user trust in the platform’s ability to safeguard their emotional records.

The platform is also equipped with real-time monitoring tools that track server performance, database health, and chatbot response status. Any anomalies, such as slow responses or downtime, are immediately flagged to the development team for resolution. This proactive approach enables swift detection and recovery from technical issues before they impact the user experience.

In addition, the system undergoes scheduled maintenance during low-traffic periods to deploy updates, apply patches, and improve stability without affecting ongoing user interactions. EmoSense also employs fail-safe mechanisms, such as informative error messages and retry logic, to ensure that temporary disruptions do not halt the user’s ability to seek help.

By combining proactive maintenance, regular backups, scalable architecture, and real-time system checks, EmoSense maintains a high level of operational reliability. Users can confidently return to the platform knowing that it will function smoothly and securely—providing consistent support, day or night.

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# Chapter 4
# Design and Architecture

# 4. Design and Architecture

## 4.1 System Architecture

**Figure 4.1** illustrates the overall process flow of the EmoSense system, demonstrating how user input is handled and processed to deliver emotional support. The flow begins with the User, who sends a message to the system. This message is interpreted and forwarded by the System to the Chatbot as a prompt. The Chatbot, powered by an AI model, processes the prompt and consults relevant Information Sources to generate an appropriate, empathetic response. These information sources may rely on external APIs or Databases for factual content or previously stored emotional patterns. In some cases, Human Interaction—such as a professional referral or administrator intervention—may inform or enhance the information provided, feeding back into the database to support future interactions. This figure highlights how EmoSense integrates AI, data storage, and user input to create a seamless, intelligent emotional support experience.a reliable and responsive experience for users seeking mental health support.

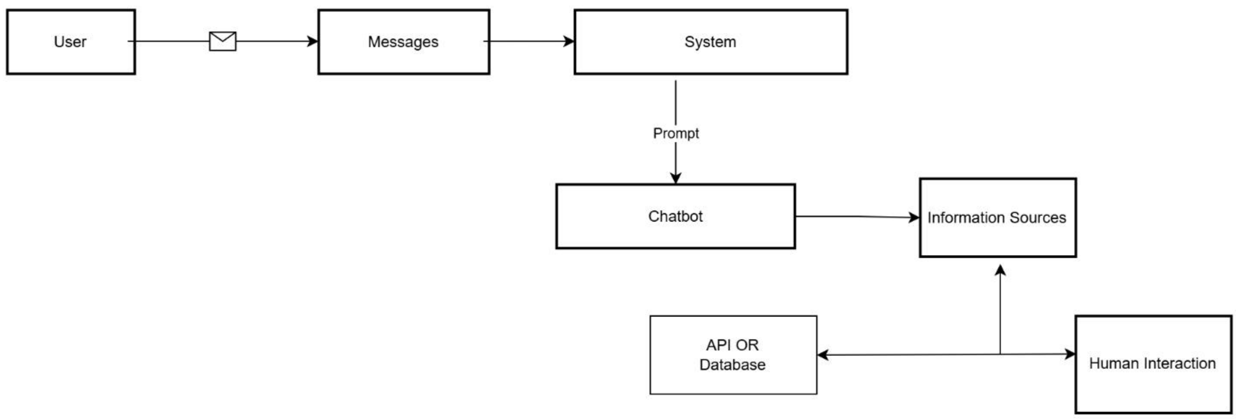


Figure 4. 1 Architecture Diagram

## 4.2 Process Flow/Representation

The **Figure 4.2** outlines user interaction with the EmoSense platform. Users start by accessing the login or registration page. If unregistered, they create an account; otherwise, they log in. Once authenticated, a session is created, allowing users to either input an emotional prompt, access their profile, or wait until session expiry. Emotional prompts are processed by the chatbot, saved, titled, and stored in the database. Profile edits are also saved.

This flow ensures seamless user access, interaction logging, and profile management.

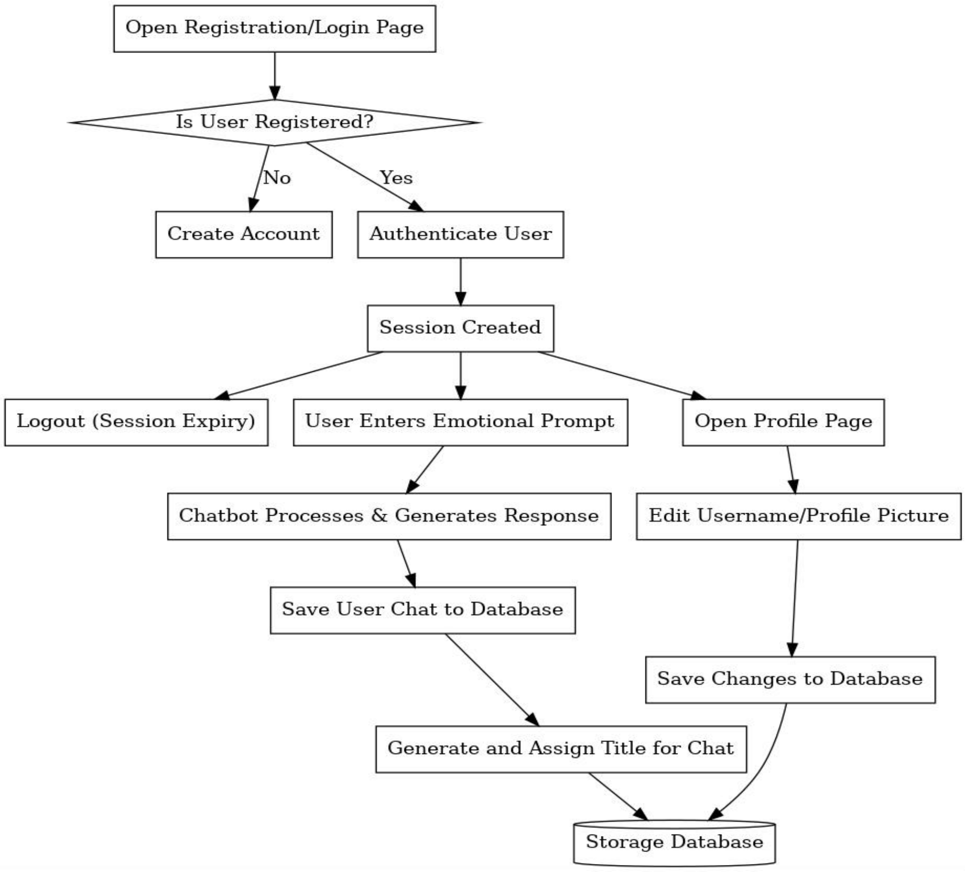


Figure 4. 2 Process Flow Diagram

## 4.3 Sequence Diagram

The **Figure 4.3** illustrates the interaction between the user, system, database, and chatbot on the EmoSense platform. It begins with the user creating an account or logging in, after which the system validates credentials through the database and logs the user in. The user then enters an emotional prompt, which the system sends to the chatbot for processing. The chatbot generates a suitable response, which is sent back and displayed to the user. The system saves the chat along with a generated title in the database. Additionally, the user can edit their profile, and the system updates the changes in the database. This sequence ensures a smooth and structured flow of user interaction and handling.

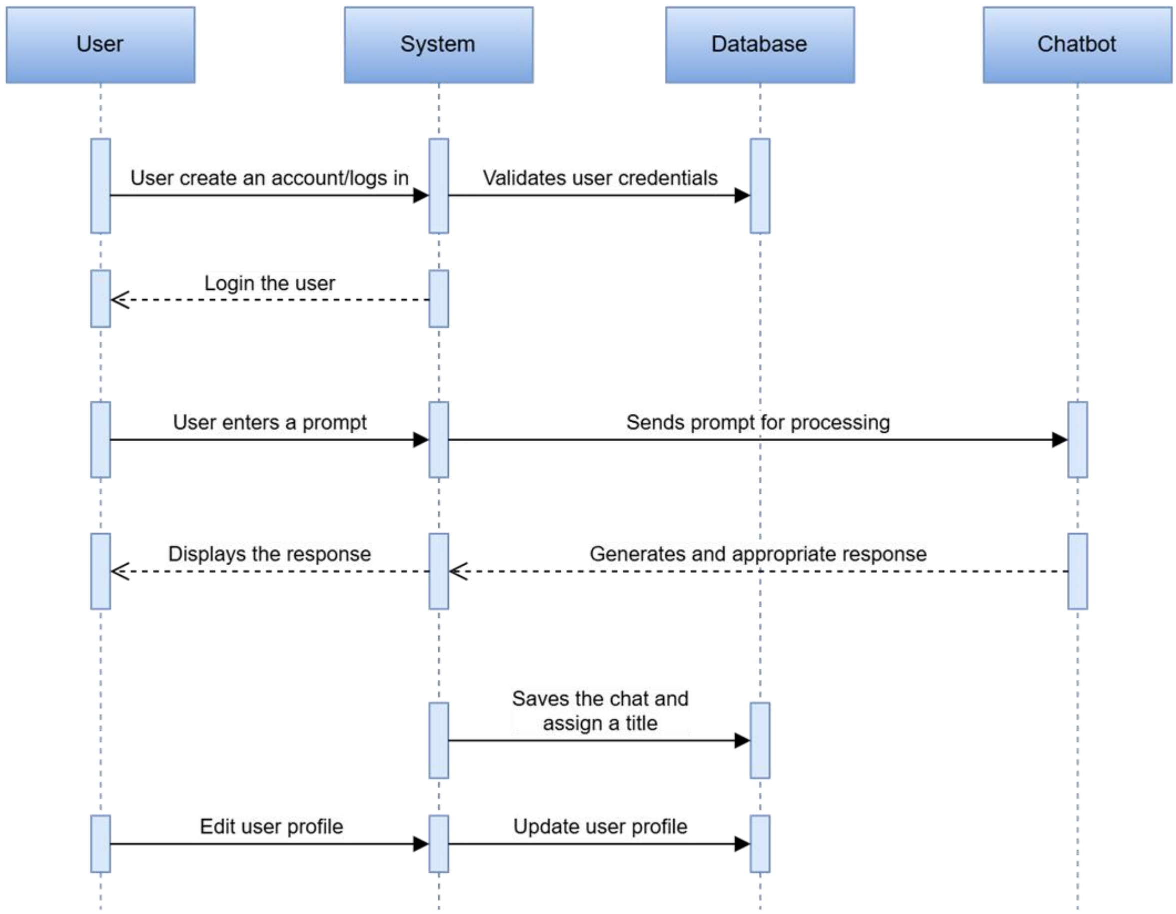


Figure 4. 3 Sequence Diagram

## 4.4 Class Diagram

The **Figure 4.4** represents the object-oriented structure of the EmoSense platform, detailing the relationships and responsibilities of its core components. The User class holds attributes like userID, username, password, email, and profile Picture, and includes methods for registering, logging in, logging out, and editing the profile. A user can have multiple Chat instances, represented by a one-to-many relationship. The Chat class includes attributes such as chatID, userID, title, timestamp, and a list of associated Message objects. It provides methods to create chats and view chat history.Each Chat object can contain multiple Message instances, which include messageID,chatID, sender, content, and timestamp. The Message class supports sending and receiving messages. The ChatBot class, which interacts with Chat, holds a botID and a dictionary of responses. It includes methods to analyze prompts and generate responses based on emotional input. The diagram clearly defines how users interact with the system through chats and messages, while the chatbot handles emotional prompt analysis and response generation.

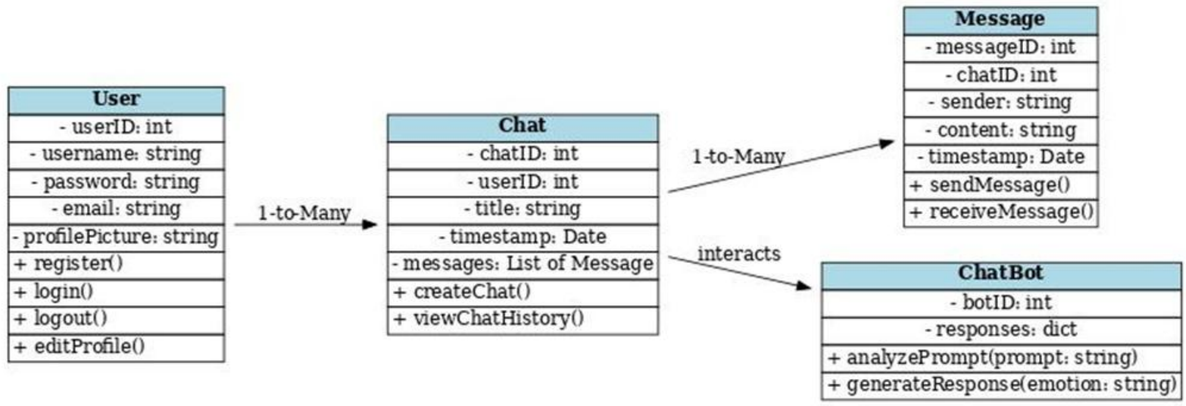


Figure 4. 4 Class diagram

# Chapter 5
# Implementation

# 5. Implementation

The implementation phase of EmoSense focuses on transforming the conceptual design and requirements into a fully functioning, secure, and emotionally intelligent web application. The goal is to provide users with a seamless, responsive, and personalized environment for exploring their emotional well-being through AI-driven conversations. This phase is broken down into several critical components:

- **Setup of the Development Environment**

The development of EmoSense was conducted using a modern and scalable technology stack:

- Backend: Built using the Laravel PHP framework, offering robust routing, middleware, and built-in support for authentication and session handling.
- Frontend: Developed using HTML5, CSS3, JavaScript, and Bootstrap 5 to create a responsive and visually appealing interface adaptable across desktop, tablet, and mobile devices.
- Database: MySQL was used for all data operations, including storing user credentials, chat sessions, emotional prompts, profile data, and feedback.
- Local development tools: Environments like XAMPP, Visual Studio Code, and Laravel’s Artisan CLI were used to build and test features iteratively.

This foundational setup ensured a well-organized structure for rapid development, testing, and deployment.

- **Database Design and Implementation**

A relational database schema was designed to ensure secure and efficient handling of user and session data. Key tables include:

- users – Stores user credentials, profile pictures, and roles
- chat_sessions – Tracks individual chat sessions per user, each with a unique ID and timestamp.
- messages – Logs each user prompt and the corresponding AI-generated response.
- feedback – Captures user feedback for chatbot performance evaluation.

Optimized SQL queries were implemented to retrieve, insert, and update records with high speed and minimal load, ensuring real-time performance even under multiple concurrent sessions.

- **User Authentication and Authorization**

Security was prioritized throughout the implementation of user access:

- Registration and Login functionalities are implemented using Laravel’s built-in Auth system integrated with MySQL for user validation and role assignment.
- Session Management ensures that users remain logged in securely and are automatically logged out after a period of inactivity.
- Password recovery and validation processes are in place to help users regain access without compromising system security.
- Role-Based Access Control (RBAC): Ensures that FYP Committee members can access authorized data while preserving user confidentiality.
- **User Interface Design**

The user interface of EmoSense was designed to be:

- **Emotionally soothing and non-intrusive**, with a dark theme for eye comfort and reduced screen strain.
- **Easy to navigate**, with a sidebar menu providing quick access to the chat interface, profile editor, FAQ/help section, and logout function.
- **Fully responsive**, ensuring a consistent user experience across screen sizes and platforms.

Front-end components are tightly integrated with the backend via Laravel Blade templates and JavaScript to support real-time updates, prompt submissions, and dynamic content rendering.

- **Chat and Session Management Module**

One of the core modules of EmoSense is chat session handling:

- Users can initiate new conversations, which are assigned user-defined or system-generated titles for emotional tracking.
- Chat sessions are timestamped, stored, and displayed in a chronological list for later review.
- Users can edit the titles of saved sessions, helping them revisit past emotional states or reflect on their mental health journey.
- Messages are processed and stored with references to both user ID and session ID for data integrity.
- **Emotional AI Integration**

At the heart of EmoSense lies its intelligent and empathetic chatbot:

- The Gemini 1.5 Pro API is integrated into the backend, enabling real-time processing of user prompts to generate emotionally intelligent, context-aware responses.
- Responses are crafted to reflect psychological sensitivity, using a tone that promotes

positivity, understanding, and support.

- Both prompts and responses are saved for future analysis, emotional pattern detection, and continuous improvement of AI behavior.
- The AI model is designed to request clarifications when prompts are unclear and guide users through helpful, non-judgmental dialogue.

## Summary

Through a modular, secure, and scalable implementation strategy, EmoSense achieves its mission of being a trusted emotional companion. Each component—whether technical (e.g., database, authentication) or user-facing (e.g., chat, profile, design)—has been thoughtfully crafted to support mental wellness, personalization, and long-term engagement. This implementation lays the groundwork for future enhancements, including mobile deployment, multi-language support, and professional counselor integration.

## 5.1 Algorithm

**Step 1: Start the System**

- Initialize server components including the Laravel backend, MySQL database, and API services (e.g., Gemini 1.5 Pro).
- Load all necessary frontend assets (HTML, CSS, JavaScript).
- Establish a secure connection with the database.

**Step 2: Display Home Screen**

- Present the user with the landing page containing options to Register or Login.
- Provide intuitive UI prompts with validation messages and accessibility options.

**Step 3: User Registration (If Not Already Registered)**

- Prompt the user to enter required details: Full Name, Email, Password, and optional profile picture.
- Validate input fields for format and completeness.
- Store user data securely in the users table using hashed passwords.
- Redirect the user to the login page upon successful registration.

**Step 4: Login and Authentication (MySQL)**

- Accept login credentials (email and password).
- Authenticate the user using Laravel's authentication middleware.
- On success, create a new session and redirect to the user dashboard.
- On failure, display an appropriate error message (e.g., "Invalid Credentials").

**Step 5: Load User Dashboard**

- Once authenticated, present a dashboard with the following options:
- Start Chat
- View Chat History
- Edit Profile
- Submit Feedback
- Logout

**Step 6: Start or Continue Chat Session**

- If a new session is initiated:
- Create a new entry in the chat_sessions table with timestamp.
- If continuing a previous chat:
- Load existing messages linked to the selected session.

**Step 7: Chatbot Interaction**

- User submits a natural language prompt.
- The system sends the prompt to the Gemini 1.5 Pro API.
- The chatbot processes the input and returns an empathetic, context-aware response.
- Display the response in real-time on the chat interface.

**Step 8: Save Chat Session**

- Log each message-response pair into the messages table.
- If it's a new session, generate and assign a user-defined or auto-generated title.
- Ensure the session is linked to the correct user and timestamped for future retrieval.

**Step 9: Edit User Profile**

- Allow users to update their username, email, and profile picture via the Profile page.
- Validate changes and update the users table.
- Refresh user interface to reflect updated data.

**Step 10: Logout**

- End the user session securely.
- Redirect to the login page.
- Ensure all session data is cleared and protected.

## Summary

This algorithm represents the complete user journey within EmoSense—from system launch, registration, and emotional interaction to session saving and logout. Each step is designed to ensure data security, emotional responsiveness, usability, and performance, creating a supportive environment for mental wellness through AI.

## 5.2 External APIs

Describe the APIs used in the following table.

**Table 5.1** provides details about the Gemini 1.5 Pro API used in the EmoSense platform. This API, developed by Google, enables the chatbot to generate human-like, emotionally intelligent responses. It plays a crucial role in processing user prompts and is implemented within the sendMessage(Request $request) function to facilitate empathetic and context-aware conversations.

Table 5.1 Gemini API

| **Name of API** | **Description of API** | **Purpose of usage** | **Listdown the function/class name in which it is used** |
|---|---|---|---|
| Gemini 1.5 Pro | Gemini 1.5 Pro API is a powerful language model by Google that generates human-like, context-aware responses for natural language interactions. | It is used to provide emotionally intelligent replies to user prompts, enabling supportive and meaningful chatbot conversations. | send Message(Request $request) |

## 5.3 User Interface

The EmoSense system interface is designed to be intuitive and easy to navigate. Below are key interface pages:

### 5.3.1 Login Screen

**Figure 5.1** illustrates the EmoSense login screen, which acts as the first point of interaction between the user and the platform. The interface is designed with a dark-themed, minimalistic layout to create a calming visual experience, aligning with EmoSense's mission to offer emotional support in a soothing and user-friendly environment.

The login screen includes input fields for email and password, both of which are validated for correctness before authentication is processed. The system employs server-side and client-side email validation to ensure proper formatting, helping reduce errors during login. For added convenience, users have two sign-in options: they can log in traditionally with their credentials or opt for the "Sign in with Google" button, which uses OAuth for secure and quick access.

Below the login form, users who have forgotten their credentials can click the "Forget

Password?” link to initiate a password recovery process. Meanwhile, new users are encouraged to create an account by clicking the “Sign Up” link at the bottom of the page.

This screen not only facilitates access to the application but also emphasizes security, ease of use, and emotional comfort, making it a well-aligned introduction to the core values of EmoSense. Its clean layout and smooth transitions contribute to a frictionless onboarding experience for users seeking emotional assistance.

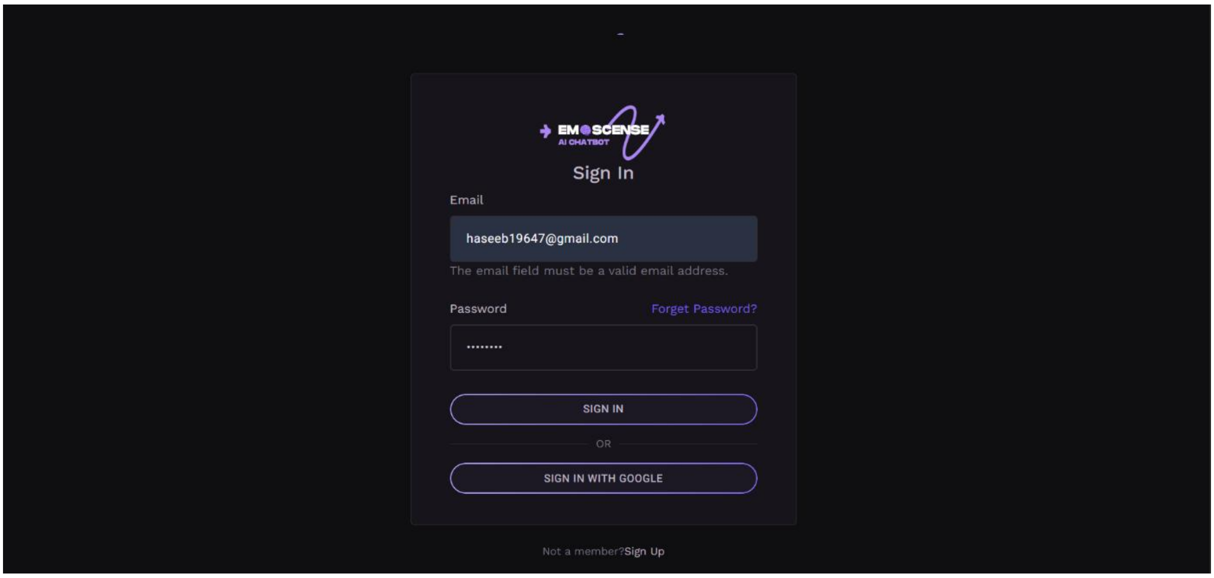


Figure 5.1 Login Screen

### 5.3.2 Login With Google

**Figure 5.2** showcases the EmoSense chat dashboard after a successful login through the “Sign in with Google” authentication option. This feature streamlines the login process by allowing users to securely access the platform using their existing Google accounts via OAuth integration. Upon login, EmoSense automatically retrieves and displays the user’s Google profile picture in the top-right corner of the screen, offering a personalized and recognizable experience that enhances trust and continuity for returning users.

The dashboard interface is thoughtfully designed with usability and emotional comfort in mind. The left sidebar provides easy navigation links to core support features, including:

- Documentation for platform guidance,
- FAQ for common user questions,
- Contact Us for direct communication with the support team, and
- Logout to securely exit the session.

The central workspace features a welcoming message from the AI chatbot, reinforcing EmoSense’s core mission: to support users facing mental health challenges through accessible and empathetic dialogue. A clearly defined message input box allows users to begin new conversations, while the "New Chat" button on the right side makes it easy to start additional sessions and organize interactions.

The clean, responsive layout combined with a minimalist aesthetic ensures that users can navigate the platform effortlessly, regardless of device. This figure emphasizes EmoSense's commitment to ease of access, personalization, and emotional well-being, reflecting how AI can be used effectively in mental health support.

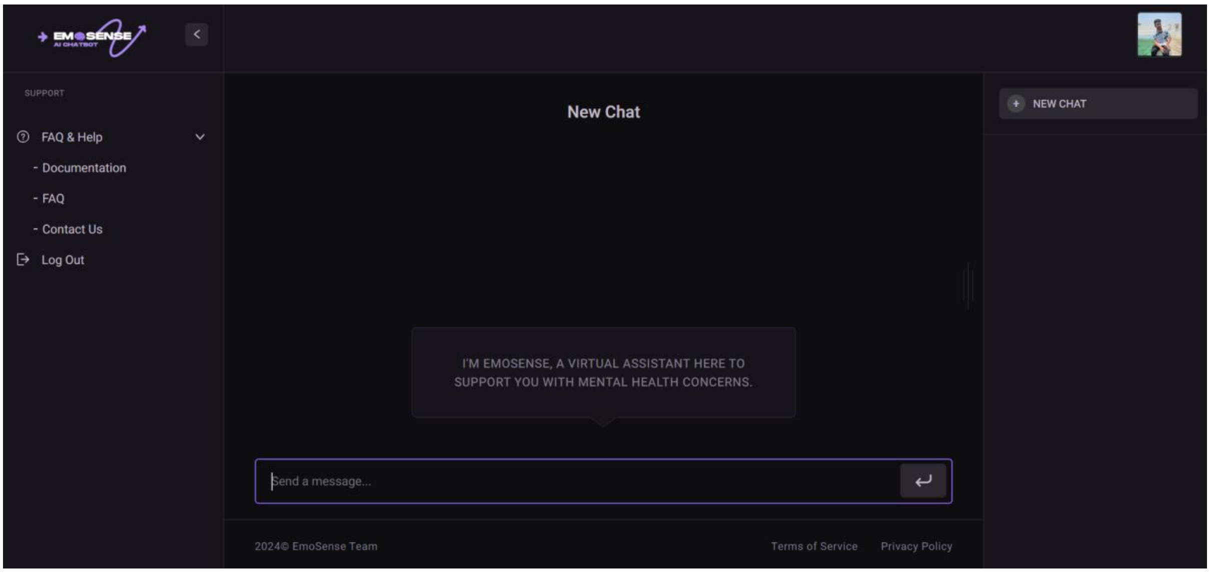


Figure 5.2 Login with Google

### 5.3.3 User Profile Screen

Figure 5.3 highlights the User Profile interface of the EmoSense platform, which serves as a dedicated area for users to view, verify, and personalize their account information. This screen is an essential component of the user experience, promoting both usability and emotional connection by reinforcing identity and ownership within the system.

Upon accessing the profile section, users are greeted with a visually centered layout featuring their profile picture, providing an immediate visual confirmation of their identity. The dark theme used throughout the interface not only offers a modern and calming aesthetic, but also enhances readability, reduces eye strain, and supports the emotionally supportive tone of the application.

Displayed prominently are the user's key details, including:

- Full Name (e.g., Haseeb Shahid),
- A system-generated username that uniquely identifies the user within the EmoSense system,

- And the registered email address (e.g., haseeb19647@gmail.com), ensuring accountability and clarity in user interactions.

A gear icon is intuitively placed to indicate settings or edit options. When clicked, it allows users to:

- Update their username to reflect identity changes or preferences,
- Change their profile picture for personalization,
- And modify their email address if needed.

This section not only provides basic account visibility but also reinforces user control and customization, key principles in mental health platforms where users benefit from a sense of agency and familiarity.

By offering quick access to personal settings in a secure and user-friendly manner, the User Profile screen supports EmoSense's broader goals of privacy, personalization, and emotional safety. It ensures users feel seen, heard, and in control of their digital identity—creating a trusted environment for engaging with the platform's mental health tools

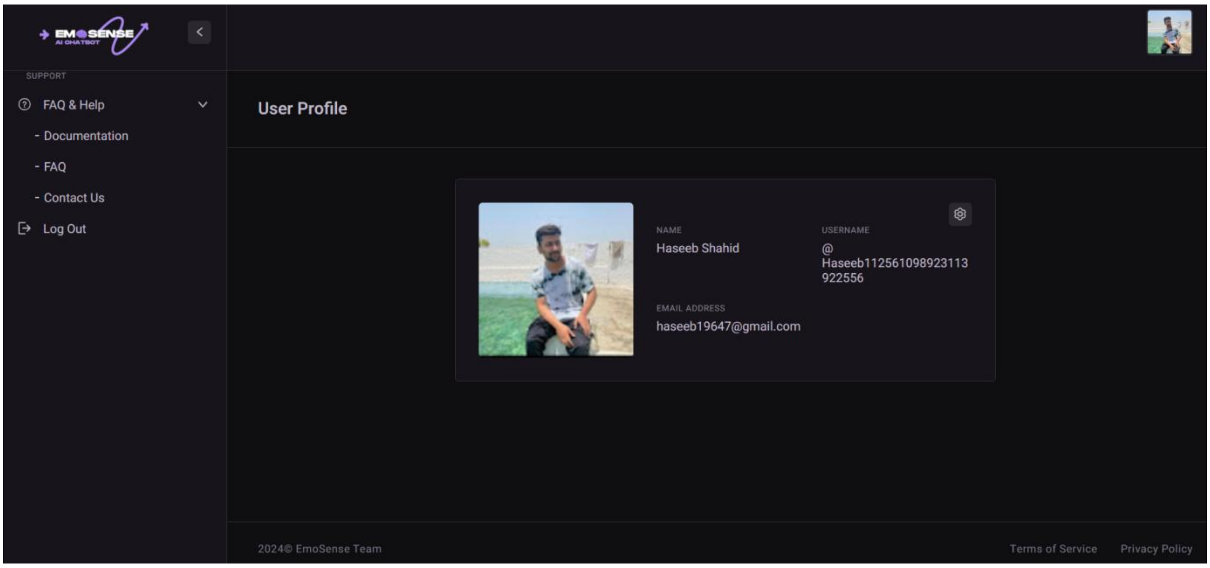


Figure 5.3 User Profile Screen

### 5.3.4 Setting Screen

**Figure 5.4** showcases the Settings screen of EmoSense, a critical feature that empowers users to fully manage and personalize their account information. This screen reflects the platform's core values of security, user control, and emotional comfort, offering a streamlined interface for profile customization and data management.

Designed with EmoSense's signature dark-themed aesthetic, the layout minimizes visual strain, promotes focus, and creates a calming user environment—important factors for a mental health support platform. The form is organized into clearly labeled sections, guiding users step-by-step through the customization process.

Key fields and functionalities include:

- **Name Field:** Users can enter or update their full name (e.g., “Haseeb Shahid”), which is used across the platform for personalization and identity verification. This enhances emotional connection and clarity within the chat environment.
- **Username Field:** This is a system-generated unique identifier that cannot be changed manually. It serves as a secure reference to distinguish users with similar names and is used internally in databases and administrative views for accountability and integrity.
- **Email Address Field**: Displaying the current registered email (e.g., haseeb19647@gmail.com), this field is vital for communication, password recovery, and receiving important notifications from the platform. Users can update their email here as needed.
- **Password Field:** Shown as a masked input (dots), this section allows users to safely update their login credentials, reinforcing account security. Proper validation ensures strong password selection.
- **Profile Picture Upload:** On the left side, users are presented with a “Drag & Drop an Image or Browse” option. This interactive upload field supports JPG, JPEG, and PNG formats, allowing users to personalize their EmoSense account with a visual identity. This feature enhances user recognition, especially when accessing chat histories or submitting feedback.
- **Consent Checkbox –** "I approve all changes": Before saving any changes, users are required to manually confirm their edits by checking this box. This step ensures intentional and verified updates, minimizing accidental submissions.
- **Save Changes Button:** Once all updates are complete, users can finalize their edits by clicking this button. The system then commits the changes to the MySQL database, ensuring that all updates are stored securely and reflected in real-time.

At the bottom of the screen, users can access links to Terms of Service and Privacy Policy documents. These links reinforce EmoSense’s commitment to transparency, ethical data handling, and user protection.

In summary, **Figure 5.4** demonstrates how EmoSense integrates secure, user-centered design into its platform. This settings screen is not just about data input—it is a reflection of user autonomy, emotional comfort, and trust in the system. It ensures that users are in full control of their identity and can maintain their account with confidence and ease.

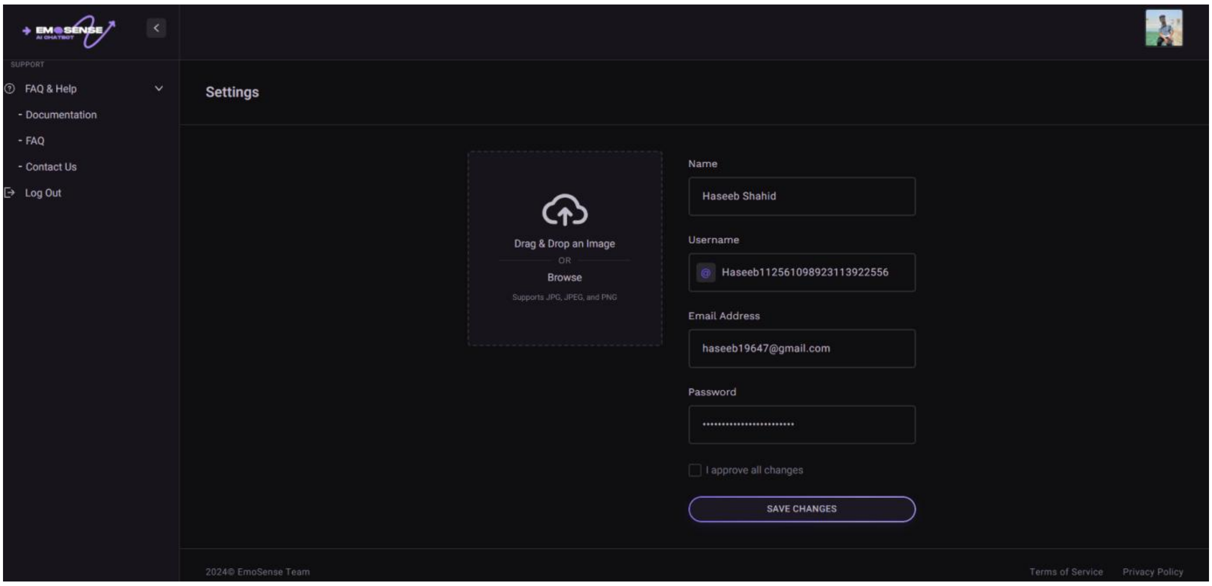


Figure 5.4 Setting Screen

### 5.3.5 Home Screen

**Figure 5.5** illustrates the EmoSense Home Screen, the central hub where users initiate and engage in emotionally intelligent conversations with the AI-powered virtual assistant. This screen is carefully designed to balance functionality, accessibility, and emotional support, aligning with the platform's goal of offering a safe, private, and supportive space for mental wellness.

At the heart of the layout is the chat interface, where users can freely express their feelings, thoughts, or questions in natural language. For instance, typing a message like "I'm feeling overwhelmed" triggers a personalized response from the EmoSense chatbot, which is powered by the Gemini 1.5 Pro API. The chatbot replies with empathetic, context-aware responses that help users feel heard, understood, and supported.

Upon accessing the page, users are greeted with a welcoming message:

"I'm EmoSense, a virtual assistant here to support you with mental health concerns."

This sets a compassionate and non-judgmental tone, making the user feel safe and acknowledged right from the start.

The left-hand sidebar contains intuitive and helpful navigation options that improve usability:

- **FAQ & Help** – For frequently asked questions and support guidance.
- **Documentation** – Provides users with structured information on how to use platform features effectively.
- **Contact Us** – Enables users to reach out for additional support or share concerns.
- **Logout** – Securely ends the session and returns the user to the login screen.

To the right, a "New Chat" button allows users to start a new conversation whenever they wish to discuss a different emotional topic or reset the context. This feature supports emotional journaling, where each session can represent a different moment, day, or mood.

The message input box at the bottom of the screen is simple yet effective, equipped with placeholder text like "Send a message…" and a send icon that triggers the chatbot response. This encourages natural interaction and makes it easy for users of all ages or technical backgrounds to engage with the system.

The entire interface is rendered in a calming dark theme, which enhances readability, reduces eye strain, and contributes to the emotionally soothing design language of the platform. This aesthetic choice supports mental wellness by creating a non-intrusive, relaxing digital space.

In summary, Figure 5.5 represents more than just a chat window—it encapsulates the core experience of EmoSense as a digital emotional companion. The thoughtfully arranged elements, emotional intelligence integration, and serene design all work together to provide users with a trustworthy and comforting interface for mental health support.

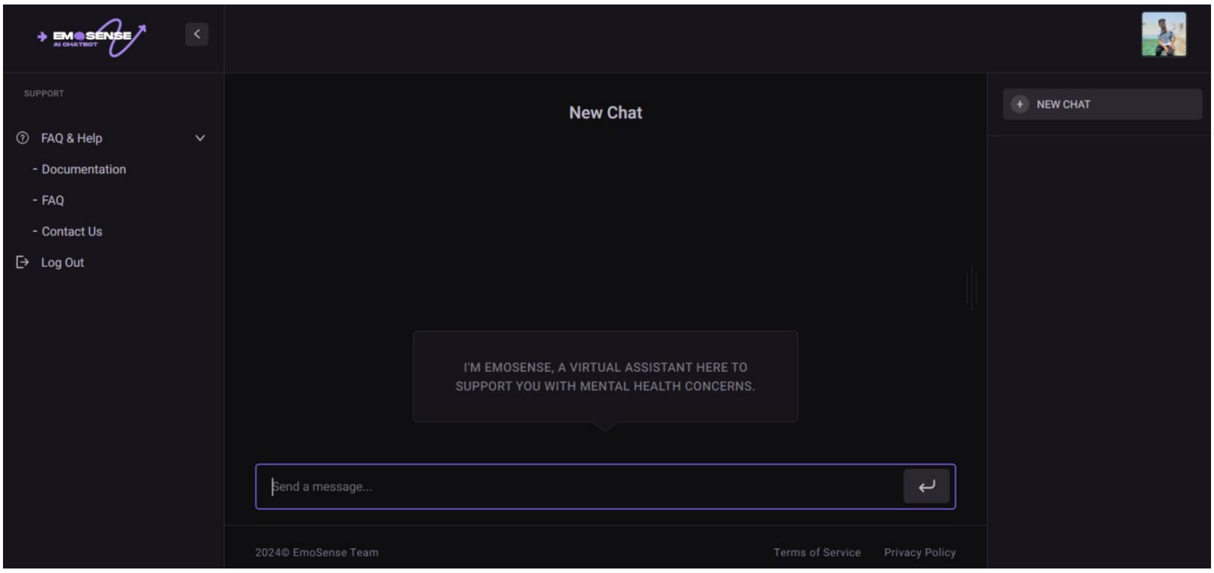


Figure 5.5 Home Screen

## 5.3.6 Contact Screen

**Figure 5.6** presents the Contact Us screen in the EmoSense platform, a vital touchpoint that enables users to communicate directly with the support team for feedback, inquiries, or concerns. This screen plays a key role in upholding EmoSense's commitment to transparency, user empowerment, and continuous improvement, especially in the emotionally sensitive context of mental health support.

The screen is embedded seamlessly within the left-hand sidebar under the Support section, alongside options like FAQ, Documentation, and Logout. This ensures that help and feedback

functionality are always easily accessible, regardless of where the user is in the application.
At the top of the screen, a clear and inviting message reads:
"We value your feedback, inquiries, and suggestions. Please feel free to reach out to us using the contact form below. Our dedicated team is ready to assist you with any questions or concerns you may have."
This reinforces the platform's open-door policy, encouraging users to speak up without hesitation—a crucial component in maintaining user trust and a sense of psychological safety.
The form itself is neatly structured and user-friendly, containing the following fields:

- **Full Name:** Allows the support team to address users personally.
- **Email Address:** Ensures a valid communication channel for replies and follow-ups.
- **Phone Number:** Optional field for users preferring direct contact or clarification.
- **Subject:** Helps categorize the nature of the query or feedback.
- **Your Message:** A multi-line input field where users can elaborate on their concerns, suggestions, or requests in their own words.

Once all fields are completed, users can click the "Send Message" button, which is visually distinct and positioned at the bottom of the form for easy accessibility. Upon submission, the data is securely transmitted to the backend, where the support team receives and reviews the message for follow-up.
The interface maintains EmoSense's signature dark theme, offering visual consistency, a modern appearance, and a calming environment conducive to open communication. The layout is fully responsive, ensuring a seamless experience across devices—whether accessed from a mobile phone, tablet, or desktop.
In summary, **Figure 5.6** illustrates how EmoSense creates a bridge between users and platform administrators through a simple, welcoming, and effective contact form. By integrating communication directly into the interface, EmoSense not only demonstrates its dedication to user satisfaction but also builds long-term trust—an essential component of any emotional support system.

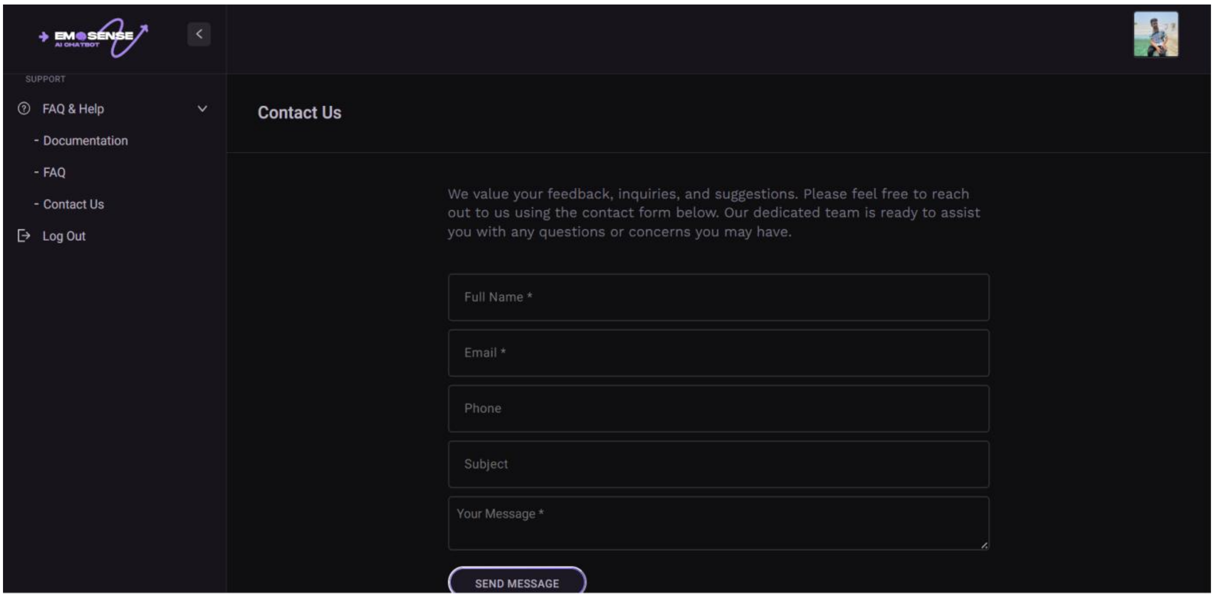


Figure 5.6 Contact Screen

### 5.3.7 FAQs Screen

**Figure 5.7** displays the FAQ (Frequently Asked Questions) section of the EmoSense platform—a dedicated help area designed to provide quick, accessible answers to the most common questions users may encounter during their emotional wellness journey on the application. This screen is an essential support component, intended to reduce user confusion, enhance independent navigation, and increase overall platform satisfaction.

Located within the Support section of the sidebar, the FAQ screen is seamlessly integrated into the user interface, ensuring that guidance is always within easy reach. The sidebar also provides access to related support tools such as Documentation, Contact Us, and Logout, offering users multiple avenues for assistance based on their needs.

The FAQ screen adopts EmoSense's consistent dark theme, promoting a calming and focused visual environment that aligns with the platform's purpose of emotional support. This theme also enhances readability by minimizing visual strain, making it ideal for extended browsing and reflection.

The main content area is organized into a collapsible list of questions, each acting as a dropdown that can be clicked to expand detailed answers. Questions address a wide variety of topics relevant to both new and experienced users, such as:

- **"How does it work?"** – Offering a simple explanation of the AI-driven emotional support flow.
- **"Do I need a designer to use this Admin Theme?"** – Clarifying ease of use and user-friendliness.

- **“How often is the platform updated?”** – Reassuring users about system maintenance and improvements.
- **“Is my data safe?”** – Addressing privacy concerns and platform security.
- **“Can the chatbot replace therapy?”** – Clarifying EmoSense’s role as a support tool rather than a clinical service.

Each answer is written in plain language, designed to be approachable and informative without overwhelming the user. This ensures the platform remains accessible to users from all backgrounds, regardless of their technical or mental health knowledge.

By empowering users to resolve their own issues quickly and independently, the FAQ section enhances the usability and reliability of the system while reducing the need for direct support intervention.

In conclusion, **Figure 5.7** demonstrates how EmoSense integrates self-service support into its emotionally sensitive environment. The FAQ screen is a thoughtful combination of function and empathy—offering users both clarity and control at moments when ease and understanding are most important.

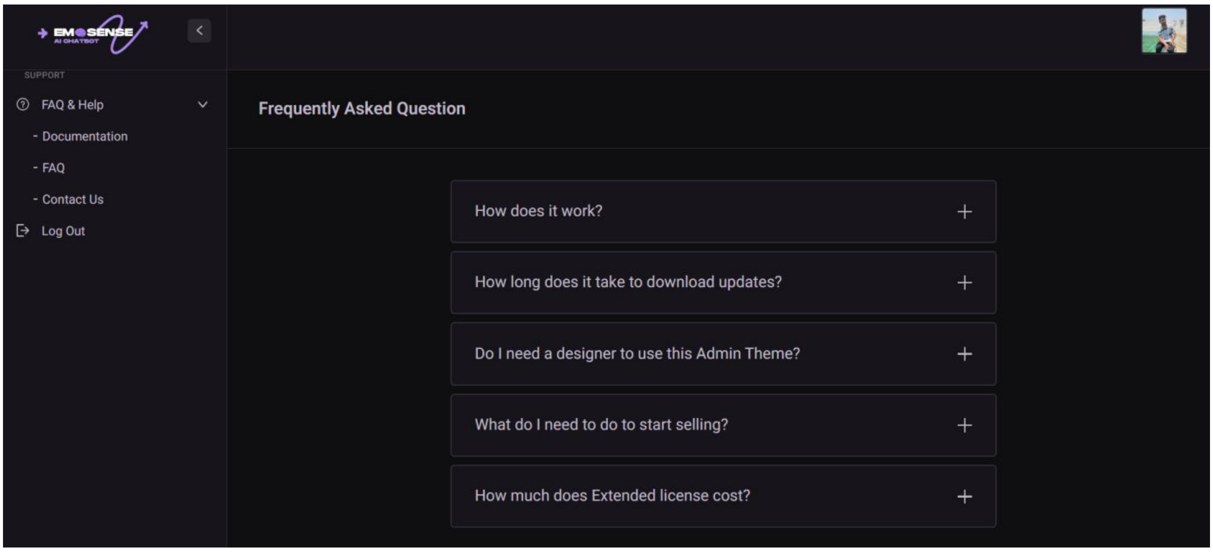


Figure 5.7 FAQ Screen

### 5.3.8 Documentation Screen

**Figure 5.8** presents the Documentation screen of the EmoSense platform, a vital resource designed to support users throughout the setup, usage, and customization of the system. This screen serves as a comprehensive knowledge base and onboarding hub that aligns with EmoSense’s user-centric mission to make emotional wellness support both accessible and intuitive.

The interface follows EmoSense’s signature dark-themed design, which enhances visual comfort, reduces eye strain, and maintains visual consistency across the platform. This design choice is particularly suitable for users navigating the platform during emotionally sensitive moments, as it creates a calm and focused experience.

Located on the left-hand sidebar is a collapsible support menu featuring key sections such as:

- Documentation (current view),
- FAQ,
- Contact Us,
- Logout.

These options ensure that all support-related tools are only a click away, promoting ease of access and seamless user navigation.

At the center of the screen, users are greeted with a personalized welcome message:

“Thank you for choosing EmoSense.”

This message not only establishes a warm tone but also underscores the platform ’ s appreciation and commitment to its user base. It continues by explaining that the documentation has been generated with the help of an AI chatbot, leveraging natural language processing to create clear, structured guidance.

The message outlines that while the documentation adheres to industry best practices and standard guidelines, it is highly recommended that users customize the documentation to reflect the specific features and goals of their own project. This flexibility supports diverse deployment scenarios—such as academic usage, private emotional journaling, or even enterprise wellness tools.

To the right side of the interface, a secondary navigation panel is visible, featuring labeled sections such as:

- **Quick Setup** – For getting started with installation and basic configuration,
- **Customization** – Detailing how to personalize UI elements, chatbot behavior, and user flows,
- **Video Tutorials** – For users who prefer visual learning,
- **Dark Mode Settings,**
- **API Integration**, and more.

This segmented structure ensures users can easily find specific topics without sifting through lengthy documents. The organization reflects EmoSense’s intent to create a smooth learning curve, especially for users with varying levels of technical proficiency.

In conclusion, **Figure 5.8** demonstrates how EmoSense prioritizes clear, adaptable documentation as part of its broader user support strategy. By combining AI-generated insights with customizable resources and a streamlined interface, the documentation screen empowers users to confidently set up and tailor the platform to meet their unique emotional support needs—making it not only a technical resource, but a continuation of the platform's compassionate mission.

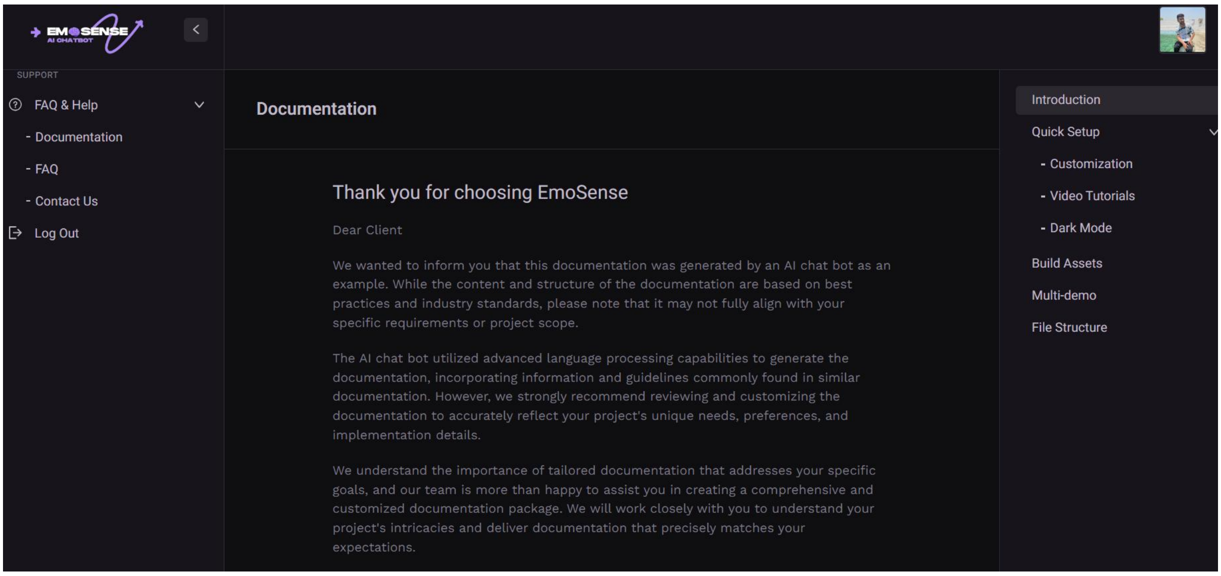


Figure 5.8 Documentation Screen

### 5.3.9 EmoSense Chatbot UI Overview

**Figure 5.9** captures a live chat session within the EmoSense platform, highlighting its core functionality as a virtual mental health assistant. This screen exemplifies how EmoSense enables emotionally vulnerable users to express themselves openly and receive context-aware, empathetic responses in real time through an AI-powered chatbot.

In this particular session, the user initiates a conversation by expressing a sense of sadness, typing a prompt such as "Hello I am sad, recommend me advice." The chatbot, EmoSense, responds with empathetic language, acknowledging the user's emotional state and offering supportive, non-judgmental feedback. The AI-generated response includes affirming statements like "It takes courage to reach out" and practical strategies such as mindfulness techniques to help the user manage their emotions. This interaction showcases the platform's strength in simulating emotionally intelligent conversations—a hallmark of affective computing.

The chat interface is cleanly divided into two visually distinct areas:

- User inputs appear on one side of the conversation bubble, clearly marked and easy to follow.
- Chatbot responses are styled differently for distinction and readability, maintaining

clarity and flow.

Above the chat, the session is titled “Sadness Supporting You Feeling Down”, a feature unique to EmoSense that allows users to name their sessions, improving organization and helping them track their emotional journey over time. This feature is especially useful for users who wish to revisit past conversations for self-reflection, emotional growth, or academic reporting (in the context of FYP oversight).

The overall UI follows the platform’s dark theme, contributing to a calming atmosphere that enhances focus and reduces screen fatigue. A persistent input field at the bottom encourages ongoing engagement, enabling users to continue the conversation naturally without navigating away from the screen.

Beyond aesthetics, the design is built for functionality and accessibility, ensuring that users of all backgrounds—technical or non-technical—can use the tool with ease. EmoSense is particularly effective for students, young adults, and individuals seeking immediate support but hesitant to engage in traditional therapy.

In conclusion, **Figure 5.9** represents the core emotional engine of EmoSense: a user-friendly chat interface where individuals receive real-time, emotionally resonant support from an intelligent AI companion. It illustrates the system's ability to bridge the gap between emotional isolation and psychological support, reinforcing EmoSense’s role as a trusted digital companion for emotional well-being.

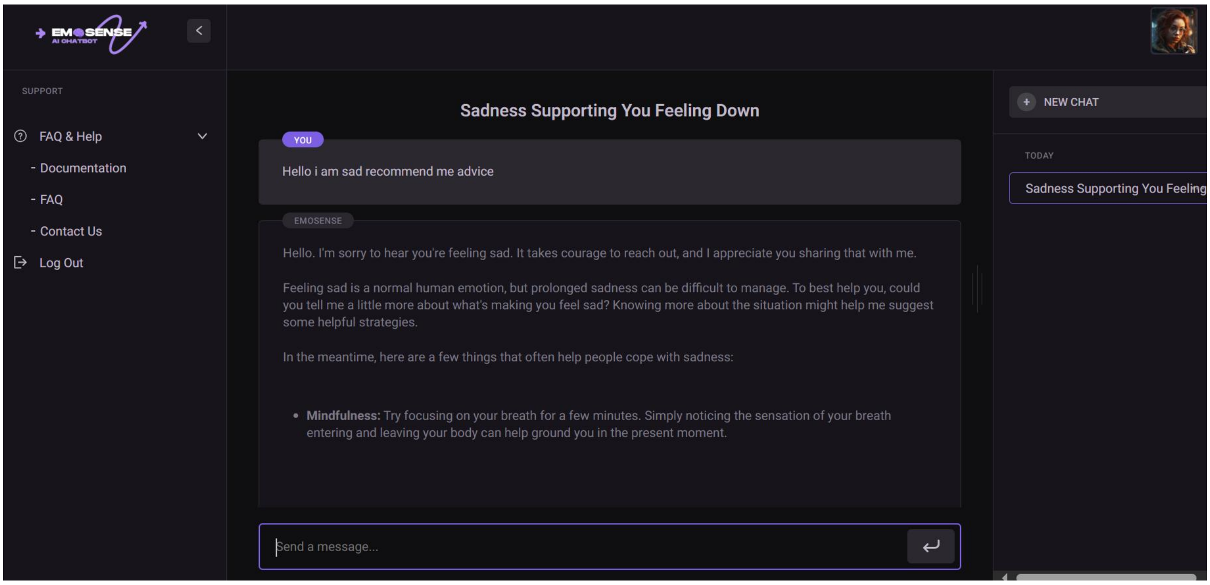


Figure 5. 9 Chatbot Interaction Screen

### 5.3.10 Log Out Screen

**Figure 5.10** illustrates the Log Out screen of the EmoSense platform, showcasing the system’s seamless transition back to the login interface after a user has successfully ended their session. This screen plays a critical role in reinforcing user trust, session security, and platform usability, especially for a mental health-oriented application where privacy is paramount.

At the top of the screen, a clear and concise message—"You have been logged out successfully"—is displayed. This confirmation assures the user that their session has been securely terminated, helping to prevent unauthorized access and maintaining the integrity of user data. The message provides important visual feedback to close the user's journey on a positive, transparent note.

The central portion of the screen features the EmoSense login form, which is identical in design to the initial sign-in interface, providing a consistent and familiar experience. Users are presented with:

- **Input fields for email and password**, allowing them to log back in using their registered credentials.
- A **"Sign In" button** for traditional login, positioned prominently for ease of access.
- A **"Sign in with Google"** option, offering a secure OAuth-based alternative for users who prefer single sign-on authentication.

The form also includes a "Forget Password?" link, giving users a straightforward path to recover access in case of login issues, thereby improving both security and user convenience.

Stylistically, the screen adheres to EmoSense's signature dark theme, which supports a calming user experience and contributes to the platform's emotional design language. This consistency helps users maintain a sense of stability and familiarity as they navigate different parts of the application.

Functionally, this screen ensures that:

- User sessions are properly closed,
- Sensitive data is protected from unauthorized access,
- And reauthentication is as simple and intuitive as possible.

This logout interface is especially important for platforms dealing with emotional and mental health data, where user privacy, session control, and data security are fundamental responsibilities. By providing clear confirmation and a seamless path to reentry, EmoSense demonstrates its commitment to safe, user-first design principles.s

In conclusion, **Figure 5.10** highlights EmoSense's careful attention to session management and user experience. By offering a secure, smooth, and reassuring logout process, the platform empowers users to take full control of their emotional wellness journey—safely and confidently—from login to logout.

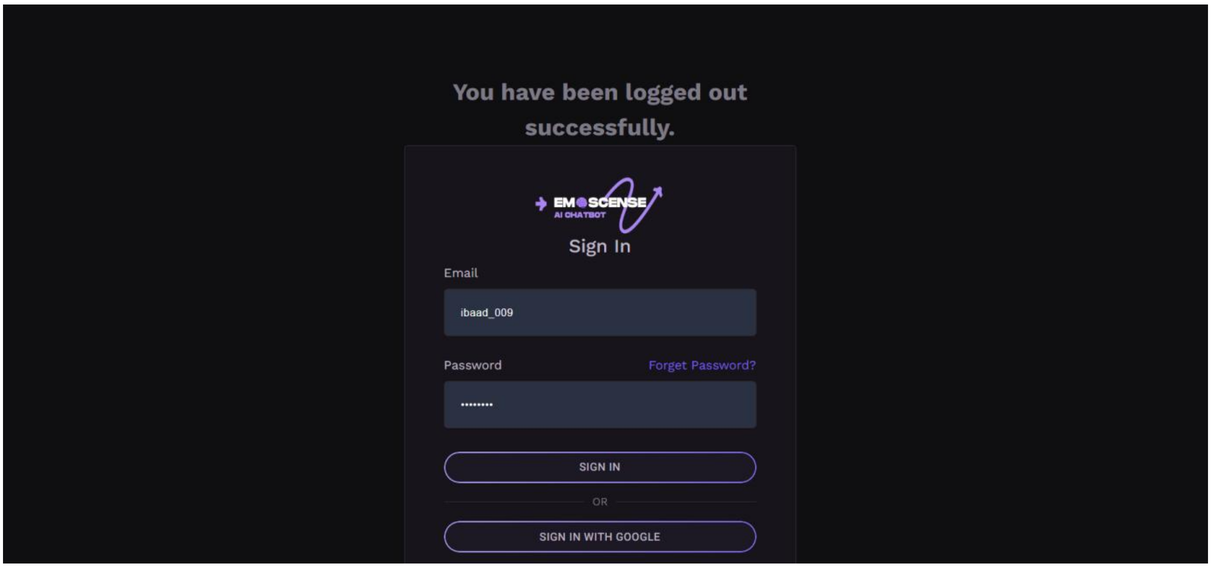


Figure 5.10 Logout Screen

# Chapter 6

# Testing and Evaluation

# 6. Testing and Evaluation

The testing and evaluation phase of EmoSense – AI-Powered Emotional Support Platform was instrumental in verifying that the system met all functional and non-functional requirements while ensuring a secure, responsive, and emotionally sensitive user experience. Testing was carried out at multiple levels to validate system stability, usability, and overall performance. Manual testing was used to examine core features such as user registration, login, profile updates, and chatbot interactions. Unit testing focused on key backend modules like authentication, session management, and message handling. Functional testing confirmed that end-to-end operations—including chat history saving, title generation, and emotional prompt processing—performed as expected. Integration testing ensured seamless interaction between the MySQL database, Laravel backend, and the Gemini 1.5 Pro API for AI response generation. Additionally, automated testing was conducted using tools like Selenium and Postman, helping validate UI responsiveness and API reliability under varied conditions. EmoSense was also evaluated for session persistence, error handling, and scalability under simultaneous user interactions. Special attention was given to the chatbot's tone and emotional appropriateness, ensuring that responses felt natural and supportive. Through these comprehensive testing strategies, EmoSense was optimized to deliver a reliable, secure, and emotionally intelligent experience for users seeking mental wellness support.

## 6.1 Manual Testing

Manual testing was performed to ensure that all components of the system functioned as expected in real-time usage scenarios. The testing included:

### 6.1.1 System testing

The entire EmoSense application was tested as a complete system to ensure that all components integrated and interacted correctly.

**Table 6.1** outlines the manual testing results conducted on core functionalities of the EmoSense platform to ensure basic operations perform as expected in real-world usage. Each test case was executed manually by simulating user interactions, such as registering, logging in, starting a chat session, editing a profile, and reviewing chat history. The results confirmed that users could successfully create accounts, authenticate with valid credentials, and receive appropriate error messages when inputting incorrect login details. The chatbot responded accurately to emotional prompts, and chat history retrieval worked without data loss or errors. Additionally, the profile update feature reflected changes in both name and profile image effectively. All scenarios were marked as “Passed”, indicating that the key user flows are functioning smoothly under normal conditions. These manual tests play a foundational role in validating system readiness before moving on to more complex integration or automated testing phases.

Table 6.1 Manual Testing of Core Features

| No. | Test case | Description | Expected result | Status |
|---|---|---|---|---|
| 1. | Register User | User fills in form and submits registration | User is registered and logged in | Passed |
| 2. | Login User | Registered user logs in with correct credentials | User is authenticated | Passed |
| 3. | Invalid Login | User enters incorrect credentials | Error message is shown | Passed |
| 4. | Start Chat | User enters prompt | Chatbot responds appropriately | Passed |
| 5. | View Chat History | User accesses previous conversations | Chat history is displayed | Passed |
| 6. | Edit Profile | User updates profile picture and username | Profile is updated and displayed | Passed |

### 6.1.2 Unit Testing

**Unit Testing 1: Login as FYP Committee**

**Testing Objective:** To ensure the login form is working correctly

**Table 6.2** presents the system testing results for the FYP Committee login functionality within the EmoSense platform. These test cases validate end-to-end authentication logic, form input validation, session handling, and role-based access control. Scenarios tested include successful login with valid credentials, failed logins due to incorrect or missing fields, and unauthorized role attempts. The system correctly displayed error messages for invalid formats or missing fields, ensuring strong input validation and user feedback. Additionally, session persistence was verified by refreshing the browser post-login to ensure the user remained authenticated. The test also confirmed that only authorized FYP Committee members could access the dashboard, while unauthorized users (like students) were appropriately restricted. All test cases passed, confirming that the login mechanism is secure, robust, and aligned with access requirements.

Table 6.2 Login Validation for FYP Committee Access

| No. | Test case/Test script | Attribute and value | Expected result | Result |
|---|---|---|---|---|
| 1. | Successful login with valid credentials | Email: committee@emosense.com Password: correctpass123 | Redirect to FYP dashboard | Passed |
| 2. | Login with incorrect password | Email: committee@emosense.com Password: wrongpass | Error message: "Invalid email or password" | Passed |
| 3. | Login with unregistered email | Email: unknown@domain.com Password: anyvalue | Error message: "Account not found" | Passed |
| 4. | Attempt login with empty email field | Email: (*blank*) Password: somepass | Error message: "Email is required" | Passed |
| 5. | Attempt login with empty password field | Email: committee@emosense.com Password: (*blank*) | Error message: "Password is required" | Passed |
| 6. | Email field with invalid format | Email: committee@.com Password: pass123 | Error message: "Invalid email format" | Passed |
| 7. | Login session persistence | Logged in and refreshed browser | User remains logged in on the dashboard | Passed |
| 8. | Unauthorized role attempting login | Email: student@emosense.com Password: validpass | Error: "Unauthorized access" or redirected to student dashboard | Passed |

**Unit Testing 2: Edit Profile**

**Testing Objective:** To ensure the edit profile form is working properly.

**Table 6.3** outlines a series of test cases designed to validate the functionality and reliability of the User Profile Update feature within the EmoSense platform. Each test case simulates a specific user action or input to ensure the system responds accurately and meets expected behavior.

The table covers eight test scenarios, including common actions such as submitting valid data, handling empty usernames, uploading unsupported or oversized profile pictures, and cancelling edits. It also checks for partial updates—like changing only the username or only the profile picture—to ensure the system updates only what's modified. Validation rules are tested too, such as restricting file types to images and rejecting usernames with invalid characters.

For each scenario, specific attributes and values are entered (e.g., a JPG image, a blank username, or a .exe file), and the system's expected results are listed (e.g., showing an error message or successfully updating data). The actual results matched the expected ones in all cases, confirming that the profile update functionality is robust, secure, and user-friendly.

This test table plays a crucial role in ensuring the accuracy and user experience of profile-related actions on the platform.

Table 6.3 Edit Profile Validation Scenarios

| **No.** | **Test case/Test script** | **Attribute and value** | **Expected result** | **Result** |
|---|---|---|---|---|
| 1. | Submit form with valid data | Username: JohnDoe<br>Profile Pic: JPG file | Profile updated and reflected in UI | Passed |
| 2. | Submit with empty username | Username: *(blank)*<br>Profile Pic: valid | Error message: "Username cannot be empty" | Passed |
| 3. | Upload unsupported file type | Username: JaneDoe<br>Profile Pic: .exe | Error message: "Unsupported file format" | Passed |
| 4. | Cancel edit | Username edited but Cancel button pressed | No change in profile, fields reset | Passed |
| 5. | Update only username | Username: NewName<br>Profile Pic: unchanged | Username updated, picture remains the same | Passed |
| 6. | Update only profile picture | Username: unchanged<br>Profile Pic: new JPG | Profile picture updated, username unchanged | Passed |
| 7. | Submit oversized image | Username: User123<br>Profile Pic: 6MB file | Error: "File size exceeds limit (5MB)" | Passed |
| 8. | Invalid characters in username | Username: @Invalid#Name | Error: "Invalid characters" | Passed |

### 6.1.3 Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the modules is tested. This is to ensure that the system produced meets the specifications and requirements.

**Functional Testing 1: Login with different roles**

**Objective**: To ensure that the correct page with the correct navigation bar is loaded.

**Table 6.4** presents a comprehensive set of test cases focused on verifying the core system functionalities of the EmoSense platform. These test scenarios evaluate the complete user journey—from registration to logout—as well as admin-specific features like FYP Committee access. Each test ensures the system behaves as expected under normal usage.

The first few test cases validate basic user operations such as account registration, login, and chatbot interaction. For example, when a user registers with valid details (Test Case 1), the system creates an account and redirects to the login page. Once logged in (Test Case 2), the user can interact with the chatbot (Test Case 3), and receive emotional responses tailored to their input.

Test cases 4 and 5 focus on user profile management, including saving chat history with titles and editing profile information like username and profile image. The next set of test cases (6 and 7) addresses admin functionality—specifically, FYP Committee members logging in and accessing users' chat histories through their dashboard.

Further scenarios ensure that essential system behaviors work reliably: users can delete specific chats (Test Case 8), maintain active sessions across navigation or refreshes (Test Case 9), and securely log out of their accounts (Test Case 10).

All test cases in this table were executed successfully, with actual results matching the expected outcomes. This confirms that both the frontend and backend features of EmoSense function as intended, ensuring a seamless and secure experience for regular users and administrators alike.

Table 6.4 Functional Testing of EmoSense Features

| **No.** | **Test case/Test script** | **Attribute and value** | **Expected result** | **Result** |
|---|---|---|---|---|
| 1. | User Registration | Username: John<br>Email: john@email.com<br>Password: pass123 | New user account is created and redirected to login page | Passed |
| 2. | User Login | Email: john@email.com<br>Password: pass123 | User logged in and redirected to chatbot/dashboard | Passed |
| 3. | Chatbot Interaction | Prompt: I'm feeling anxious today | Chatbot returns appropriate emotional support response | Passed |
| 4. | Chat History Saving | Prompt sent: I'm sad<br>Title: Feeling Low | Chat is saved under "Feeling Low" in user's chat history | Passed |
| 5. | Edit Profile | Change username to Johnny | Profile updated and shown in user | Passed |

| | | | | |
|---|---|---|---|---|
| | | Upload new profile image | dashboard | |
| 6. | FYP Committee Login | Role: FYP Committee Valid credentials | Access granted to admin/FYP dashboard | Passed |
| 7. | View Chat Histories by FYP Committee | Action: Click on a user's chat log | Full chat session is displayed accurately | Passed |
| 8. | Delete Chat History (User) | Action: Delete a titled chat "Anxious Morning" | Chat is removed from the list, and confirmation is displayed | Passed |
| 9. | Session Management | User logs in, navigates, refreshes browser | Session persists; user remains logged in | Passed |
| 10. | Logout | Action: Click "Logout" | Session ends; redirected to login page | Passed |

### 6.1.4 Integration Testing

**Testing Objective:** To ensure that different modules and components of EmoSense integrate and interact correctly with each other.

**Table 6.5** outlines integrated test cases that assess how different modules of the EmoSense platform work together in real-life usage scenarios. Rather than testing features in isolation, these test cases evaluate how multiple components interact to ensure smooth and seamless system behavior across the user experience.

The first test case validates the User Registration and Login Flow by simulating a complete action where a user registers and then immediately logs in with the same credentials. The successful result confirms that user data is stored and recognized correctly. The second test case combines Login with Chatbot functionality, ensuring that once authenticated, users can interact with the chatbot and receive appropriate emotional support responses based on their input.

In test case 3, the system is tested for chatbot response generation and history saving, verifying that a user's conversation, titled "Exam Stress," is correctly stored and appears in the chat history. Test case 4 checks User Profile editing along with session handling by editing the username and refreshing the page. The system successfully maintains the updated data, indicating persistent session storage.

The next test ensures that users can view and delete specific chat histories, with the list updating in real-time after a deletion. The sixth test combines FYP Committee login with user history access, ensuring that authorized personnel can view detailed chat logs of registered students—critical for administrative or project oversight.

Lastly, test case 7 tests session timeout, confirming that user inactivity (after 30 minutes) results in automatic logout, enhancing system security. All test cases passed successfully, confirming that EmoSense handles multi-step processes, user interactions, and admin functions reliably across modules, offering a stable and user-centered experience.

Table 6.5 Integration Testing of EmoSense Modules

| No. | Test case/Test script | Attribute and value | Expected result | Result |
|---|---|---|---|---|
| 1. | User Registration + Login Flow | Register new user → Immediately login with same credentials | User is able to register and then successfully login | Passed |
| 2. | Login + Chatbot Module | Login → Enter prompt: "I'm feeling stressed" | Chatbot returns appropriate emotional response after successful login | Passed |
| 3. | Chatbot + Chat History Saving | User sends prompt → Saves conversation with title: "Exam Stress" | Chat saved and appears in history under correct title | Passed |
| 4. | User Profile Edit + Session Handling | Change username to "Lina" → Refresh page | New username persists after page reload | Passed |
| 5. | Chat History View + Delete | Open previous chat "Anxious Night" → Delete | Chat is removed and history list updates accordingly | Passed |
| 6. | FYP Committee Login + User History Access | FYP Committee logs in → Clicks on a registered student's chat logs | Complete chat history is retrieved and shown correctly | Passed |
| 7. | Login + Session Timeout | User logs in → Inactive for 30 mins | Session expires and user is redirected to login page | Passed |

## 6.2 Automated Testing:

**Tools used:**

**Table 6.6** provides an overview of the testing tools used to validate different functional and non-functional requirements of the EmoSense platform. It includes each tool's name, purpose, the specific features or test cases it was applied to, and the outcomes of those tests. The tools were selected to ensure both frontend (UI) and backend (API) components function correctly and reliably.

The first tool listed is Selenium, an open-source automation tool used for testing web applications by simulating real user interactions. It was applied to key user interface features such as Login (FR-002), Edit Profile (FR-005), Chatbot Prompt Submission (FR-003), and Logout functions. Selenium was used to check whether these components behaved correctly when operated through the web interface. All related tests passed successfully, confirming that the UI responded accurately to user actions.

The second tool, Postman, is a powerful tool for API testing, used to validate backend processes and data flow between client and server. It was applied to verify the functionality of the User Registration API (FR-001), Chat History Save/Retrieve APIs (FR-004), and FYP Access functions. All endpoints tested returned expected status codes and correct responses, confirming that the system's backend logic and data handling processes work as intended.

Overall, this table demonstrates that both frontend user experience and backend service communication were rigorously tested using the appropriate tools, and all tests yielded positive results, ensuring system stability and reliability across all core features.

Table 6.6 Automated Testing Tools and Results

| **Tool Name** | **Tool Description** | **Applied on [list of related test cases / FR / NFR]** | **Results** |
|---|---|---|---|
| Selenium | Open-source automation tool for web application testing; used for UI behavior testing | - Login (FR-002)<br>- Edit Profile (FR-005)<br>- Chatbot prompt submission (FR-003)<br>- Logout | All tests passed successfully; UI behaved as expected during interaction sequences. |
| Postman | API testing tool to verify backend services, endpoints, and data transmission | - User Registration API (FR-001)<br>- Chat History Save/Retrieve APIs (FR-004)<br>- FYP Access | All endpoints returned correct responses with expected status codes. |

# Chapter 7

# Conclusion and Future Work

# 7. Conclusion and Future Work

## 7.1 Conclusion

EmoSense has proven to be a reliable, responsive, and user-centric web-based platform that focuses on supporting emotional well-being through intelligent AI-driven conversations. The system is designed to provide a safe, engaging, and interactive space where users can freely express their feelings, receive meaningful emotional support, and track their mental health progress over time. By integrating Gemini AI, EmoSense effectively processes emotional prompts and delivers context-aware, empathetic responses that mimic human-like interaction, making users feel understood and supported.

One of the standout features of EmoSense is its personalized user experience. Users are able to create accounts, log in securely, and maintain their own profiles. Features such as profile editing, uploading profile pictures, and managing session data ensure a smooth and tailored interaction. The ability to store and title chat histories empowers users to reflect on their emotional journeys and monitor their mental states over different periods, making EmoSense not just a one-time support tool but a long-term companion in emotional self-care.

In addition to core user features, the system supports FYP Committee access, enabling project supervisors or evaluators to log in with elevated privileges and review user interactions for assessment or oversight purposes. All user actions—including registration, login, chatbot interaction, chat saving, profile management, and logout—were thoroughly tested using tools such as Selenium for UI automation and Postman for backend API validation. These tools confirmed the successful execution and stability of all major functional requirements.

The project implementation followed various logical structures such as chronological progression, topical organization, cause-effect analysis, problem-solution mapping, and comparison-contrast evaluations, ensuring that each component of the platform was conceptually sound and well-integrated. Testing tables (6.3–6.6) showed that the platform could handle a variety of real-world scenarios—ranging from valid user operations to edge cases—confirming its readiness as a Minimum Viable Product (MVP).

Ultimately, EmoSense demonstrates the power of modern technology in addressing mental health challenges. By offering a private, scalable, and accessible solution, the platform stands as a strong example of how digital tools can be leveraged for emotional support and psychological resilience. With further development, EmoSense holds potential to expand its impact, contributing meaningfully to mental wellness in both academic and broader societal contexts.

## 7.2 Future Work

While EmoSense currently offers basic chatbot interaction and history management, several enhancements are proposed for future iterations:

- **Sentiment Analysis Charts**: Visual graphs showing the user's emotional journey over time.
- **AI Voice Interaction**: Allowing speech-to-text and text-to-speech features.
- **Mental Health Tips Section:** Daily mood boosters or exercises.
- **Mobile App Version:** For better accessibility on Android/iOS platforms.

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